Cosmology

Core track, 2009 Winter Legacy Cosmology

Lectures by Leonard Susskind

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About These Notes

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Chapter 1

Geometry of the Expanding Universe

The point is the expanding universe and its geometry. The aim is to describe what it means for space to expand before asking what drives the expansion. Newtonian $F = ma$ cosmology comes later, and only after that do geometry and dynamics get assembled into the full Friedmann–Robertson–Walker framework.

1.1 Comoving Coordinates on an Expanding Line

Start with the simplest possible example: one-dimensional space, not spacetime. Imagine an infinite line with equally spaced galaxies on it. The coordinate x does not yet measure a physical distance. It labels the galaxies. One may think of x as the name of a galaxy, and that name stays attached to the galaxy even if the whole arrangement stretches or contracts.

Let the physical distance from the galaxy called $x = 0$ to the galaxy called $x = 1$ be a. Then the physical distance between any two galaxies is

$$d = a\,\Delta x. \tag{1.1}$$

The coordinate difference Δx is only the difference between labels. The factor a converts that label difference into an actual distance measured by meter sticks.

Now imagine the line as an elastic thread being stretched. The labels stay with the galaxies, but the distance between neighboring galaxies changes with time. The same relation becomes

$$d(t) = a(t)\,\Delta x. \tag{1.2}$$

At this stage homogeneity means simply that neighboring galaxies are equally spaced everywhere, even though that common spacing may vary with time.

1.2 Hubble's Law and the Absence of a Center

To find the relative velocity of two fixed galaxies, keep the pair fixed so that Δx does not change with time. Differentiating gives

$$\begin{aligned} \dot{d} &= \dot{a}\,\Delta x && (1.3)\\ &= \frac{\dot{a}}{a}\,a\,\Delta x && (1.4)\\ &= \frac{\dot{a}}{a}\,d. && (1.5) \end{aligned}$$

If we define

$$H(t) \equiv \frac{\dot{a}}{a}, \tag{1.6}$$

then

$$\dot{d} = H(t)\,d. \tag{1.7}$$

This is Hubble's law.

The historical phrase "Hubble constant" is misleading. The quantity is constant in space, not in time. At a given moment it is the same for every galaxy, but in general it changes as the universe evolves. It is better thought of as the Hubble parameter.

The law says that more distant galaxies recede faster. It does not single out a center. Nothing in the argument privileges $x = 0$, or any other point. Every observer sees the same rule. The expanding universe in this model is therefore not a blast from one distinguished location.

Redshift enters immediately. If two galaxies exchange light signals while moving apart, the received light is reddened. The greater the recession speed, the greater the redshift. Spectral lines therefore tell us about recession speeds. Distances are harder to measure, and Hubble's early estimates were rough. His first plot was not a perfect straight line but a scattered set of points with large uncertainties. Even so, the geometric principle survived the noise: recession speed is proportional to distance.

Question from the audience. If the galaxies are far enough apart, does Hubble's law make them move faster than the speed of light?

[a] **Response.** Yes, in this recession sense it does. What you are saying is correct. We will come back to that for sure. It is one of the most important things.

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[a] Lecture timestamp: 00:20:23.

1.3 Higher Dimensions, Isotropy, and the Time Dependence of $a(t)$

Nothing essential changes when more spatial dimensions are added. In two dimensions imagine a square grid of galaxies labeled by (x, y), and in three dimensions by (x, y, z). The coordinates are again comoving labels. They move with the galaxies instead of measuring physical distance directly.

Why should the spacing in the x-direction be the same as the spacing in the y-direction? The answer is observational. If galaxies were systematically more crowded in one direction than another, the sky would look different in different directions. On sufficiently large scales it does not. That directional sameness is isotropy.

In two dimensions the distance between two galaxies is

$$d = a\sqrt{(\Delta x)^2 + (\Delta y)^2}, \tag{1.8}$$

and in three dimensions

$$d = a\sqrt{(\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2}. \tag{1.9}$$

For a fixed pair of galaxies the coordinate differences remain constant, so differentiating again gives

$$\dot{d} = \frac{\dot{a}}{a}\, d. \tag{1.10}$$

Hubble's law is unchanged.

Homogeneity means sameness from place to place. Isotropy means sameness from direction to direction. Neither is exact on small scales. Looking through the Milky Way is different from looking out of its plane, and individual galaxies are not uniformly spread at short distances. But after averaging over many galaxies, the large-scale universe looks approximately the same everywhere and in every direction.

Question from the audience. If velocity is proportional to distance, and the distance is increasing, does that mean a fixed pair of galaxies must be accelerating away from each other?

[a] **Response.** No. Hubble's law says that at a given moment the recession velocity is proportional to the distance. It does not by itself say whether the velocity of one fixed pair increases or decreases with time. That depends on the detailed time dependence of $a(t)$. If $a(t) \propto t$, neighboring galaxies separate at constant speed. If $a(t) \propto t^2$, the recession speed of a fixed pair increases with time. If $a(t) \propto \log t$, it decreases with time. The law $\dot{d} = (\dot{a}/a)d$ is always true as a function of distance, but acceleration and deceleration are separate questions.

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[a] Lecture timestamp: 00:40:22.

For a long time cosmologists expected the universe to be decelerating, since gravity should slow an initial expansion the way gravity slows an object thrown upward. Here the point is only that Hubble's law by itself does not determine the behavior of $\dot{a}$ or $\ddot{a}$.

Question from the audience. How do you set the initial value of the scale factor a?

[a] **Response.** Its overall normalization is arbitrary. What matters physically is not the numerical size of a by itself, but how it changes. If one defines another scale factor $b = 3a$, then $\dot{b}/b = \dot{a}/a$. Rescaling a by a constant changes conventions or units, not the Hubble parameter.

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[a] Lecture timestamp: 00:46:18.

1.4 From the Spatial Metric to the Spacetime Metric

Now replace Δx by dx. The physical distance between neighboring points is

$$ds = a\, dx, \tag{1.11}$$

so

$$ds^2 = a^2\, dx^2. \tag{1.12}$$

This is the spatial metric in one dimension. The coefficient of dx^2 is

$$g_{xx} = a^2. \tag{1.13}$$

An expanding universe can therefore be described by saying that the spatial part of the metric depends on time.

To pass from space to spacetime, use the relativistic interval. In one spatial dimension,

$$d\tau^2 = dt^2 - a^2(t)\, dx^2. \tag{1.14}$$

For a comoving observer, $dx = 0$, so $d\tau = dt$. Ordinary cosmic time is the proper time measured by clocks moving with the galaxies.

The blackboard picture at this point is useful because it puts the comoving worldlines, the line element, and the metric notation together.

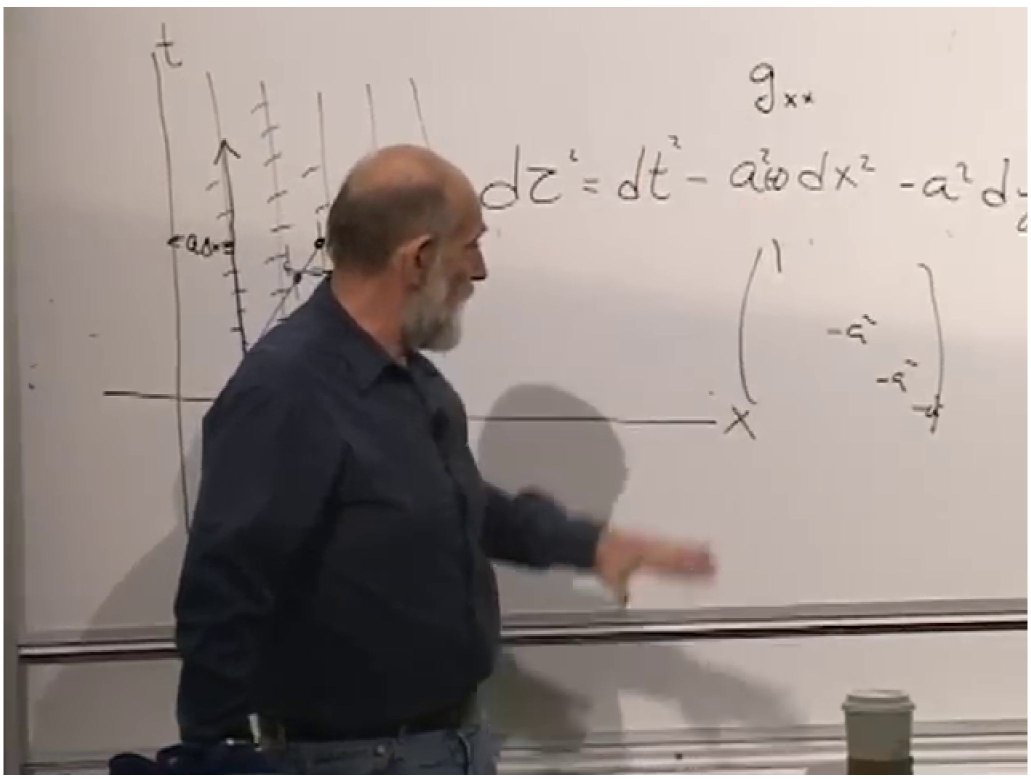

Figure 1.1: Comoving worldlines in an expanding one-dimensional universe together with the corresponding spacetime interval.

In three spatial dimensions the same pattern gives the cautious extension[1]

$$d\tau^2 = dt^2 - a^2(t)\big(dx^2 + dy^2 + dz^2\big), \tag{1.15}$$

with diagonal metric

$$g_{\mu\nu} = \mathrm{diag}\big(1, -a^2, -a^2, -a^2\big). \tag{1.16}$$

[1]The board is partly hidden around 00:56:38–00:58:51. This three-dimensional form follows the spoken extension from one spatial dimension to three together with the visible diagonal pattern on the board.

The off-diagonal terms vanish,

$$g_{tx} = g_{ty} = g_{tz} = 0. \tag{1.17}$$

Homogeneity means that the metric coefficients may depend on time but not on spatial position. Isotropy means that the three spatial directions enter in the same way.

Question from the audience. Are we not supposed to think of dx as a distance?

[a] **Response.** No. dx is not the distance. $a\,dx$ is the distance. dx is just a label interval. A useful way to think about it is to imagine that, at some moment, someone draws a coordinate mesh through the galaxies and then requires the mesh to follow them. The coordinate separation stays fixed by construction, while the physical distance changes. One could put the units of distance into x instead of into a; that is a convention. What matters is the product $a\,dx$.

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[a]Lecture timestamp: 01:01:18.

Question from the audience. The metric coefficient of dt^2 is equal to 1. Is that just by convention also?

[a] **Response.** Yes. There is no physics in that by itself. If the coefficient depended only on time, one could change the time variable so that it became just dt^2. Likewise one may choose units with $c = 1$. What is not a convention here is that the coefficient does not depend on position, although it may depend on time.

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[a]Lecture timestamp: 01:03:24.

Question from the audience. If you put this into Einstein's field equations, what do you get for the energy-momentum tensor?

[a] **Response.** That is the next kind of question, but it belongs to dynamics and not to the present discussion. One can put this metric into Einstein's equations and ask what energy-momentum distribution would make it a solution. That is very important, but for the moment the subject is geometry, not dynamics.

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[a]Lecture timestamp: 01:05:59.

1.5 What the Horizon Hides

The large-scale universe is treated as homogeneous and isotropic only as an approximation. It is certainly not homogeneous on the scale of stars, planets, or even individual galaxies. It is not exactly isotropic in every local observation either.

A farmer's field gives the right scale dependence. On one scale it looks regular; on a much smaller scale it plainly does not; on a scale larger than the whole field it again ceases to look homogeneous. Cosmology works the same way. After averaging over many galaxies, the large-scale distribution looks roughly uniform. On sufficiently small scales it does not.

The universe is bigger than the part we can see, and bigger than the part we can ever see. On the scales that can actually be probed, it is isotropic and homogeneous apart from local variations. Beyond that there is no direct observational guarantee.

Question from the audience. You said we will never really directly find out. Does that imply that there might be an indirect way to find out?

[a] **Response.** Perhaps. Most thinking these days is that on big enough scales the universe is not homogeneous, and my own bet is the same. But it is a bet I will never win and never lose, because for basic fundamental reasons we will not be able to see much farther than we can see now. That is the existence of a horizon.

[a] Lecture timestamp: 01:10:34.

Question from the audience. Couldn't you use gravity to look for masses beyond what you can see?

[a] **Response.** No. A distant mass can simply put us into free fall. Sitting in a closed room, we are in the gravitational field of the Sun and orbiting it, but without seeing outside there is no easy way to tell that the Sun is there. Only tidal forces reveal it, and if the source is far enough away those tidal effects can be too small to detect.

[a] Lecture timestamp: 01:11:53.

The horizon is therefore an ultimate barrier to direct knowledge. He even says, with virtual certainty, that the universe is at least a thousand times larger in volume than the region we can ever see, and leaves the reason for that claim for later.

1.6 Bounded Without Boundary

Now come to closed geometries. The infinite line is open and unbounded. A line segment is bounded in the sense of having only a finite extent. Bounded does not mean a boundary, but finite extent.

A cube with walls will not do. It violates homogeneity, because somebody near a wall is not equivalent to somebody far from a wall, and it violates isotropy, because even somebody at the center sees different things in different directions.

The interior of a circle or ball also fails. Somebody at the center sees the same thing in every direction, but that symmetry belongs only to one special point. Move away from the center and the wall is near in one direction and far in another. The space is then neither homogeneous nor isotropic for generic observers.

On very large scales the working assumption is that the universe is homogeneous and isotropic. This is the cosmological principle. It is useful, but it is not protected from revision.

Question from the audience. Does isotropy everywhere imply homogeneity?

[a] **Response.** Yes. If every observer sees the same thing in every direction, that implies homogeneity. But the converse is false. Homogeneity by itself does not imply isotropy. A universe can be homogeneous and still have a preferred direction, for example if it were filled with arrows all pointing the same way.

[a] Lecture timestamp: 01:18:57.

The right bounded model is the surface of a featureless sphere. A two-dimensional creature confined to that surface finds every place locally like every other place and every direction like every other direction, even though the world is finite. That is bounded without boundary.

1.7 A Closed One-Dimensional Universe on a Circle

To keep the algebra simple, reduce the sphere example to one spatial dimension: life on a circle. In one dimension isotropy means that left and right are equivalent. A circle has exactly that feature. Choose one point and call it $\theta = 0$. Every other point is labeled by an angular coordinate θ, measured in radians. A point just to one side of the origin can be described either by a small negative angle or by an angle just short of 2π. That periodicity is part of the geometry.

The inhabitants of the circle have access to the circumference, not to an external radius or center. They can measure the total distance around by marching along the circle until they return to the same point. If one draws the circle in an ambient plane, one may also introduce a radius-like quantity, but that belongs to the embedding picture, not to the intrinsic geometry they measure.

The blackboard picture is useful here because it keeps the angular labels and the total circumference on the same drawing.

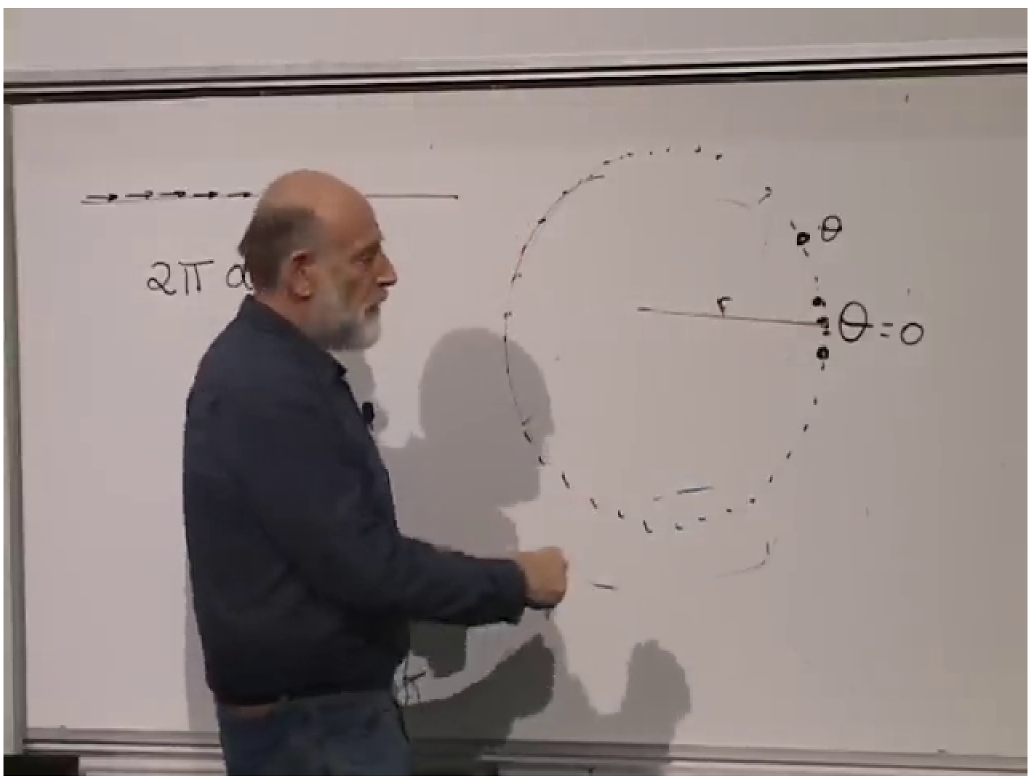

Figure 1.2: Angular description of a closed one-dimensional universe, with total circumference $2\pi a$.

A convenient convention is[2]

$$L = 2\pi a, \tag{1.18}$$

where L is the total circumference.

For a small angular separation $\Delta\theta$, the physical distance is

$$d = a\,\Delta\theta. \tag{1.19}$$

[2]Around 01:26:40–01:29:20 the notation shifts briefly between taking the whole circumference itself as the scale factor and writing it as $2\pi a$. The later convention is used here because it is the one carried into the arc-length formula.

Passing to infinitesimals gives

$$ds = a\,d\theta, \tag{1.20}$$

and therefore

$$ds^2 = a^2\,d\theta^2. \tag{1.21}$$

This is the local metric on the circle.

If a depends on time, the closed one-dimensional universe expands or contracts. For a fixed pair of galaxies,

$$d(t) = a(t)\,\Delta\theta, \tag{1.22}$$

so

$$\dot{d} = \dot{a}\,\Delta\theta = \frac{\dot{a}}{a}\,d. \tag{1.23}$$

Hubble's law survives here as well, at least for pairs that are not so far apart that the other way around the circle becomes ambiguous.

The spacetime metric becomes

$$d\tau^2 = dt^2 - a^2(t)\,d\theta^2. \tag{1.24}$$

If a is constant, the spacetime picture is a cylinder. If $a \propto t$, it is a cone. If $a(t)$ changes more irregularly, the picture becomes vase-like.

Run the picture backward and the galaxies get closer and closer together. If there were 360 galaxies around the circle, they would be well spread out at late times, very close together earlier, then closer still. Eventually they would not be separate galaxies anymore, and earlier still not even separate atoms. But there is still no special point in space.

That is the cleanest answer to the question of where the Big Bang happened in the model. As $a(t) \to 0$, every separation $a(t)\,\Delta\theta$ goes to zero. The whole universe shrinks everywhere at once. The galaxies are not flying away from one privileged location in an external space. The geometry itself is changing scale. In that sense the answer is: everywhere.

Question from the audience. You said the spacetime is curved. Is it curved if that were a cylinder?

[a] **Response.** No. If it were a cylinder, it would not be curved. That is the case where a is constant in time. It looks curved only because it is drawn in a higher-dimensional space. If one cuts it open, it can be laid flat without stretching.

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[a] Lecture timestamp: 01:38:04.

From there the discussion turns to the cone. If $a \propto t$, the spacetime picture is a cone. Away from its tip the cone is also flat in the same intrinsic sense: it can be opened up into a plane. The tip is different. It is a conical singularity.

The next step is the geometry of the spherical universe and then the dynamics that determine how $a(t)$ grows or shrinks.

Chapter 2

Newtonian Cosmology and the Friedmann Equation

We were talking last time about the metric of space. The next question is dynamics: how does the scale factor change with time? First comes a little geometry, and then Newtonian cosmology. The Friedmann equation will not yet be derived from general relativity. It will be derived from Newton's equations.

2.1 Homogeneity, Isotropy, and a Time-Dependent Metric

Homogeneity and isotropy are the standing assumptions. If the universe is homogeneous and isotropic and it changes with time, then Hubble's law is always true. At any fixed time the recession velocity is proportional to distance,

$$v = H(t)\, d. \tag{2.1}$$

The proportionality is linear. What is constant is not time, but space: the same proportionality factor applies everywhere at a given instant.

Question from the audience. Does that mean linearly proportional?

[a] **Response.** Yes. At a fixed time the recession velocity is linearly proportional to distance. The proportionality factor is the Hubble constant. In general it depends on time, but it does not vary from point to point in space.

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[a]Lecture timestamp: 00:02:06.

That is why the phrase "Hubble constant" is dangerous. It is constant with respect to position in space. It is not constant with respect to time.

A natural question comes next. If space expands, why do meter sticks not expand too? The answer is that meter sticks are held together by forces enormously stronger than anything associated with the cosmological expansion on atomic scales. Bound objects stay bound. What expands is the distance between sufficiently disconnected things, such as galaxies far enough apart that they do not hold each other together strongly.

Question from the audience. Is there any issue about the fact that the measuring rods are expanding too?

[a] **Response.** No. The measuring rods are not expanding. The coordinates are expanding. The forces which hold meter sticks together are far stronger than the effects due to the expansion, so ordinary bound objects do not expand with the background. Galaxies far enough apart do separate, but meter sticks do not.

[a]Lecture timestamp: 00:04:00.

One can even imagine a tiny extra repulsive effect between atoms. That still would not tear a rod apart. It would only shift the equilibrium separation a little bit, the way a small extra pull changes the equilibrium position of a stiff spring.

Question from the audience. You can't model it though as a force between points, is that correct? Because otherwise it'd be an acceleration.

[a] **Response.** A small extra force between atoms can be imagined, but if it is tiny compared with the binding forces it only shifts the equilibrium position slightly. Expansion by itself does not create an ordinary force law. Some kinds of expansion can be modeled in terms of a force, namely accelerated expansion, but that is a later question.

[a]Lecture timestamp: 00:05:29.

So the useful coordinates are coordinates embedded in the expanding matter distribution. Galaxies with no extra peculiar motion keep the same coordinate values. What changes with time is the relation between coordinate separation and physical distance measured in meters. That relation is codified by the scale factor $a(t)$.

For flat space the spatial metric may be written

$$ds^2 = a(t)^2 \left(dx^2 + dy^2 + dz^2\right). \tag{2.2}$$

Here a is the actual distance, measured in meters, between neighboring points separated by one unit of coordinate distance. If the universe expands, a becomes time-dependent.

To pass to spacetime, start from the special-relativistic form

$$d\tau^2 = dt^2 - ds^2, \tag{2.3}$$

and substitute the spatial metric. This gives

$$d\tau^2 = dt^2 - a(t)^2 \left(dx^2 + dy^2 + dz^2\right). \tag{2.4}$$

The coefficients do not depend on position in space. They depend only on time. That is the statement of homogeneity in metric language. The diagonal metric has the form

$$g_{\mu\nu} = \mathrm{diag}\left(1, -a(t)^2, -a(t)^2, -a(t)^2\right). \tag{2.5}$$

[1]

There are no off-diagonal terms. No $dx\,dy$, no $dx\,dt$, no other mixed terms. The metric is almost as simple as that of flat spacetime. The peculiarity is that the relation between spatial measure and time is itself time-dependent.

[1]Editorial clarification: The full diagonal matrix written here follows the spoken metric-tensor description around 00:14:33. The board frame shows the line element clearly, but not a fully legible matrix.

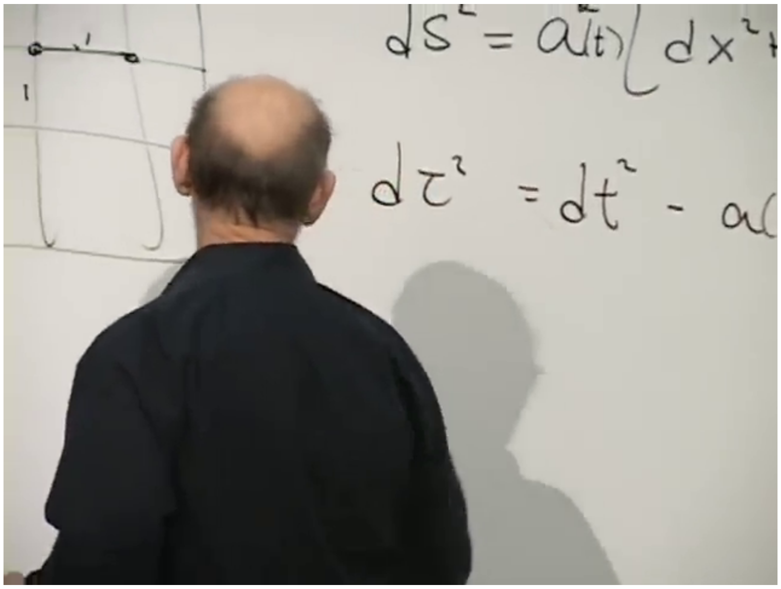

Figure 2.1: Comoving grid sketch and spacetime interval with a time-dependent scale factor.

The picture carries the point. Alice and Bob remain one unit apart in the comoving coordinates, but that one coordinate unit represents more and more meters as time goes on. If $a(t)$ were zero at some time in the past, then every physical separation would be zero. Everything would be squashed on top of everything else. That is why $a \to 0$ corresponds to infinite density. If a grows, the universe becomes more and more diffuse.

This form of the metric is not valid in every frame. It is the form in the frame moving with the galaxies.

Question from the audience. In that equation, the small t, $a(t)$, whose t is that?

[a] **Response.** It is the time measured by clocks moving with the galaxies. Imagine identical clocks made in the same cosmic clock factory and carried by the galaxies. In this frame the time variable is exactly what those clocks measure. Along a galaxy worldline the proper time is the same as this coordinate time.

---

[a]Lecture timestamp: 00:19:19.

Indeed, along a comoving galaxy worldline,

$$dx = dy = dz = 0, \tag{2.6}$$

so

$$d\tau^2 = dt^2, \tag{2.7}$$

and therefore

$$dt = d\tau. \tag{2.8}$$

Question from the audience. Are there transformations that leave this form of the metric invariant?

[a] **Response.** Rotations do, because $dx^2 + dy^2 + dz^2$ is invariant under rotations. But Lorentz boosts do not preserve this form. Matter itself picks out the frame moving with the galaxies, and in some other frame the world would not look isotropic.

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[a]Lecture timestamp: 00:23:08.

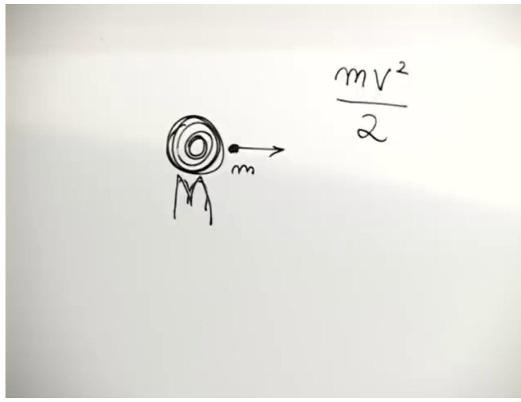

Figure 2.2: Outgoing test mass and the kinetic-energy term $\frac{1}{2}mv^2$.

2.2 A Projectile in a Gravitational Field

Before applying Newton to the whole universe, it helps to step back to a simpler problem: a heavy mass M, a much smaller mass m, and an initial outward impulse. After the impulse, the small mass simply moves in the gravitational field and does whatever Newton says it should do.

For the cosmological discussion, the important thing is not the full orbit problem. It is energy conservation. The kinetic energy is

$$\frac{1}{2}mv^2. \tag{2.9}$$

The gravitational potential energy contains the product of the masses, Newton's constant, and one power of the distance in the denominator. In the board writing the factors appear in the order mMG/D, so it is best to keep that order here:

$$-\frac{mMG}{D}. \tag{2.10}$$

Thus the total energy is

$$\frac{1}{2}mv^2 - \frac{mMG}{D}. \tag{2.11}$$

Energy conservation says only that this quantity does not change with time for the particular particle under discussion. Different particles may have different energies.

The negative sign matters. Gravitational potential energy is negative because gravity is attractive. Bringing the particle closer creates a larger amount of negative potential energy. Moving it farther away makes the potential energy less negative.

Question from the audience. The potential energy increase with distance rather than decrease?

[a] **Response.** Yes. Forget the constants and look at the graphs of $1/D$ and $-1/D$. Force pushes toward decreasing potential energy. An attractive force therefore corresponds to the upside-down curve, $-1/D$, so the potential energy increases with distance and becomes more negative as the particle moves inward.

[a] Lecture timestamp: 00:34:13.

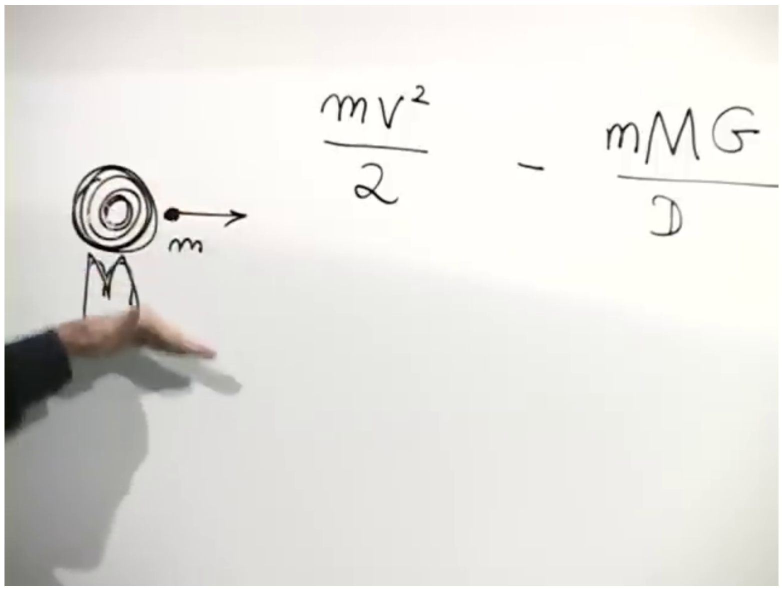

Figure 2.3: Total Newtonian energy for a radially moving test mass: kinetic term minus gravitational potential.

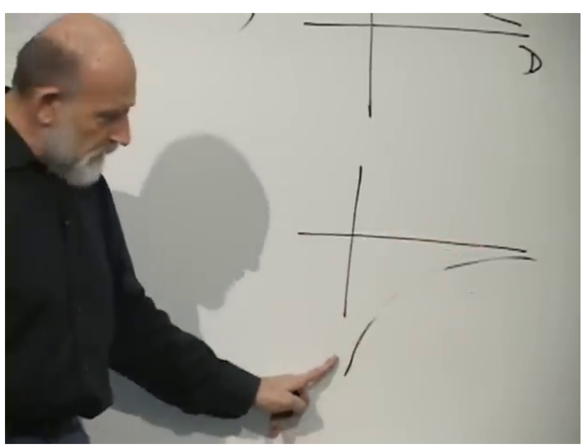

Figure 2.4: Qualitative comparison of $1/D$ and $-1/D$ for the sign of gravitational potential energy.

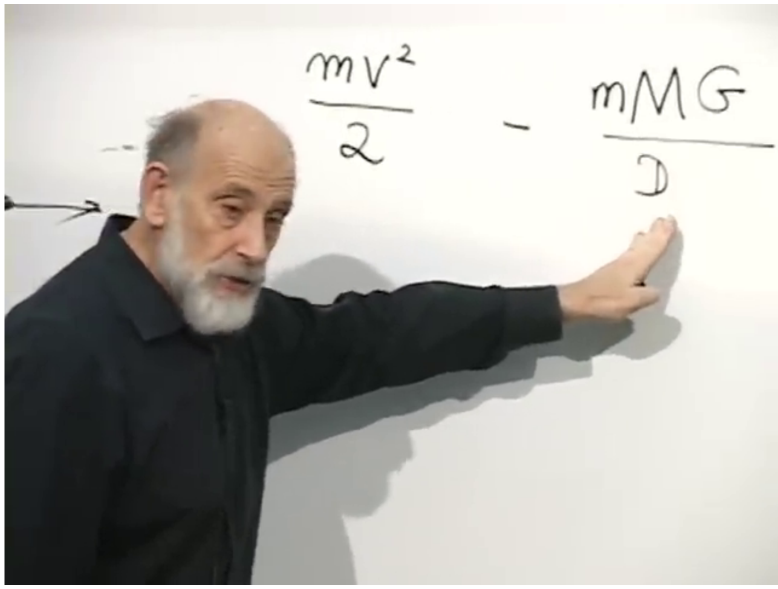

Figure 2.5: At small outward speed, the negative potential term dominates the total energy.

2.3 Escape Velocity

Now classify the possibilities. The energy can be positive, negative, or zero. Those are the only three cases.

- If the energy is negative, the particle is bound. Throw it up with too small a velocity and it rises, slows, turns around, and falls back.
- If the energy is positive, the particle escapes. Gravity decelerates it at first, but not enough to stop it, and eventually it moves off with a nonzero asymptotic outward velocity.
- If the energy is exactly zero, the particle is at the escape velocity. That is the marginal case between escaping and not escaping.

For sufficiently small initial velocity, the potential term dominates from the start and the energy is negative.

In the zero-energy case the distance keeps increasing, but the speed keeps decreasing. The particle cannot come to rest at any finite time. If it did, it would fall back in. So the distance is unbounded, while the velocity tends toward zero.

This is the direct analogy that matters for cosmology. The universe can behave the same way. If the initial outward Hubble flow is large enough, everything escapes from everything else. If it is too small, gravity wins and there is a giant crunch. There is also the marginal case, exactly analogous to escape velocity.

2.4 Newton's Theorem

The cosmological reduction works because of Newton's theorem for the inverse-square law.

If matter is distributed spherically symmetrically about some center, then the gravitational force on a particle outside the distribution is exactly the same as if all the mass were concentrated at the center. But the theorem is stronger than that. If the particle is inside the distribution, then only the mass at smaller radius contributes. The mass at larger radius gives no net force.

The cleanest version of the theorem is the thin shell. A perfect spherical shell exerts no gravitational force on a particle anywhere inside it, even away from the center. That is surprising at first sight, but it is true for the $1/r^2$ law.

Question from the audience. In the example you gave there, should we be a little surprised that it seems like as the particle moves across the boundary, the force is discontinuous?

[a] **Response.** Yes. For an ideal infinitely thin shell it is surprising, and it is special to the $1/r^2$ force law. A real shell has thickness, so the variation is smoothed out, but the sharp statement for an ideal shell is exactly the remarkable property of Newton's theorem.

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[a]Lecture timestamp: 00:48:19.

A picture then makes the theorem plausible. Replace a point mass by a point source of water. In three dimensions the outward flow decreases like $1/r^2$. Put such sources continuously all around a spherical shell. Outside the shell the flow looks exactly as if everything were concentrated at the center. Inside the shell there can be no net flow, because there is no source at the center. The same structure is what underlies the gravitational theorem.

The point here is not to rederive the theorem. It is to use it.

2.5 A Homogeneous Universe as a One-Body Problem

Now go back to cosmology. Take a little patch of the world, filled with a uniform gas of galaxies. One may put oneself at the center if one likes. That is only a convenience. Anybody can call themselves at the center, and the same equation must come out.

Choose a galaxy on the x-axis at comoving coordinate x. Since the universe is isotropic, the choice of axis does not matter. Draw a sphere centered on $x = 0$ and passing through that galaxy. By Newton's theorem, the force on the galaxy is exactly the same as if all the mass inside that sphere were concentrated at the center, while everything outside contributed nothing.

The sphere expands with the galaxies. It always contains the same galaxies, and therefore the same total mass, even though its physical radius changes with time.

The motion of that galaxy therefore obeys exactly the same kind of energy equation as the projectile:

$$\frac{1}{2}mv^2 - \frac{mMG}{D} = \text{constant}. \tag{2.12}$$

Here the M now means the mass inside the expanding sphere, not the central mass in the warm-up picture. The algebra is then simplified by multiplying by 2 and dividing by m. Since a constant divided by m is still just a constant, one may write

$$v^2 - \frac{2MG}{D} = \text{constant}. \tag{2.13}$$

Again, "constant" means constant in time for that particular galaxy.

Now relate this to the scale factor. The physical distance to the galaxy is

$$D = ax, \tag{2.14}$$

because x is a fixed comoving coordinate. Its velocity is

$$v = \dot{D} = \dot{a}\,x. \tag{2.15}$$

The mass inside the sphere is the density times the volume. If ρ is the mass density, then

$$M = \frac{4\pi}{3}\rho D^3 = \frac{4\pi}{3}\rho a^3 x^3. \tag{2.16}$$

Substituting into the energy equation gives

$$\dot{a}^2 x^2 - \frac{2G}{ax}\left(\frac{4\pi}{3}\rho a^3 x^3\right) = \text{constant}, \tag{2.17}$$

or

$$\dot{a}^2 x^2 - \frac{8\pi G}{3}\rho a^2 x^2 = \text{constant}. \tag{2.18}$$

Both terms on the left-hand side are proportional to x^2. Everything else there is the same everywhere. Therefore the right-hand side must also be proportional to x^2. That is not a contradiction with calling it a constant. It means constant with respect to time for each fixed x, but it may still scale with x^2 from one galaxy to another. That makes sense. The more distant galaxies are moving faster.

So divide away the common factor of x^2. The remaining constant is now independent of both time and space. Write it in the historically standard way as $-k$:

$$\dot{a}^2 - \frac{8\pi G}{3}\rho a^2 = -k. \tag{2.19}$$

After dividing by a^2,

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho - \frac{k}{a^2}. \tag{2.20}$$

This is the famous Friedmann-Robertson-Walker equation, though the lecture also remarks that it is really Friedman's equation. The sign convention is historical. Writing $-k$ does not mean k must be positive. k may be positive, negative, or zero. In units with the speed of light set equal to 1, k is dimensionless.

2.6 Density and the Matter-Dominated Equation

The next question is immediate: what is ρ? It is not a constant. As the universe expands, the density decreases.

To express that dependence, take a unit cube in coordinate space: one unit on a side in x, y, and z. The same galaxies remain inside that cube for all time because the cube expands with the coordinates. So the mass inside that coordinate cube is fixed.

[2]

[2] Editorial clarification: The lecture reuses the symbol M at several stages. Here $M_\square$ is used only to keep the mass in a unit coordinate cube distinct from the earlier central mass and sphere-enclosed mass.

The physical side length of the cube is a, so its physical volume is a^3. Therefore

$$\rho = \frac{M_\square}{a^3}. \tag{2.21}$$

This is the matter-dominated case: the right-hand side involves only the density of ordinary, nonrelativistic matter.

Question from the audience. According to that equation, the Hubble constant is a function of ρ. Therefore it's not a constant.

[a] **Response.** Exactly. The Hubble constant is not a constant. It depends on the density, and the density changes with time. The lecture even jokes that the final examination could consist of one question: Is the Hubble constant a constant?

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[a]Lecture timestamp: 01:19:21.

Substituting $\rho = M_\square/a^3$ into Friedmann's equation gives

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G M_\square}{3a^3} - \frac{k}{a^2}. \tag{2.22}$$

At this point the arbitrary choices are clear. One needs the mass in a unit coordinate cube and the value of k. The three possible signs of k correspond to the three energy cases of the projectile problem.

- If $-k > 0$, the energy is effectively above escape velocity, and the universe expands forever.
- If $-k < 0$, gravity eventually wins and the universe recollapses.
- If $k = 0$, the universe is exactly at the escape velocity.

Question from the audience. Is $k = 0$? Then the velocity between two galaxies is approaching zero?

[a] **Response.** Yes. In the critical case the velocity approaches zero, but the distance is still unbounded. The galaxies keep separating, only more and more slowly. If they actually came to rest at any finite time, they would fall back in.

---

[a]Lecture timestamp: 01:21:12.

2.7 The Critical Case $k = 0$

Now choose the simplest case, $k = 0$. That is the exact escape-velocity case, and it is simpler only because the mathematics is simpler. The equation becomes

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G M_\square}{3a^3}. \tag{2.23}$$

The right-hand side is always positive and never zero. Therefore $\dot{a}/a$ cannot pass through zero. The universe cannot turn around in this case. It is either always expanding or always shrinking. Take the expanding branch.

Now guess a power law,

$$a(t) = ct^p. \tag{2.24}$$

Then

$$\dot{a} = cp\, t^{p-1}, \tag{2.25}$$

so

$$\frac{\dot{a}}{a} = \frac{p}{t}. \tag{2.26}$$

Substituting into the differential equation gives

$$\frac{p^2}{t^2} = \frac{8\pi G M_\square}{3c^3 t^{3p}}. \tag{2.27}$$

For the powers of t to match, one must have

$$3p = 2. \tag{2.28}$$

So

$$p = \frac{2}{3}. \tag{2.29}$$

Then the constants must satisfy

$$\frac{4}{9} = \frac{8\pi G M_\square}{3c^3}, \tag{2.30}$$

hence

$$c^3 = 6\pi G M_\square. \tag{2.31}$$

Therefore

$$a(t) = (6\pi G M_\square)^{1/3}\, t^{2/3}. \tag{2.32}$$

There can still be an additive time shift. It is absorbed by replacing t with $t - t_0$. Once one chooses the origin of time so that $a = 0$ at $t = 0$, that remaining constant is gone.

So the universe expands as the two-thirds power of time,

$$a(t) \propto t^{2/3}. \tag{2.33}$$

It does not expand linearly with time. Linear growth would be the analog of unaccelerated motion. Here the expansion slows down.

If one pushes the formula backward to $t = 0$, the scale factor goes to zero and the density blows up. Saying $a = 0$ is extreme. We do not know what happens to matter when it becomes too dense. The real point is that at some early stage everything was so crowded that atoms, nuclei, and quarks were all packed together in an extraordinarily dense state.

2.8 The Hubble Constant and the Age of the Universe

In the critical model,

$$H(t) = \frac{\dot{a}}{a} = \frac{2}{3t}. \tag{2.34}$$

So the Hubble constant is approximately one over the age of the universe,

$$t = \frac{2}{3H}. \tag{2.35}$$

This is one of the main payoffs of the calculation. Measure the Hubble constant today, and one gets an age estimate for the universe, at least back to the point where the Newtonian model ceases to make sense.

The quoted age is roughly 13 to 14 billion years. The text also says that $k = 0$ is observationally very realistic: to the level of precision quoted there, no significant deviation from $k = 0$ is known.

Question from the audience. I'm just sort of confused about what M and a are.

[a] **Response.** Both contain arbitrariness from the choice of grid. Change the fiducial box and the amount of matter in it changes, and the numerical value of the scale factor changes too. What is physical is the density, M/a^3, or equivalently combinations such as $a/M^{1/3}$ that do not depend on how the grid is chosen. In the $k = 0$ solution the unknown constant multiplying $t^{2/3}$ cancels out of $\dot{a}/a$, so the age relation $t = 2/3H$ does not depend on the fiducial box mass.

---

[a]Lecture timestamp: 01:39:16.

That cancellation is immediate in the solution. If

$$a(t) = Ct^{2/3}, \tag{2.36}$$

then

$$\dot{a} = \frac{2}{3}Ct^{-1/3}, \tag{2.37}$$

and therefore

$$\frac{\dot{a}}{a} = \frac{2}{3t}. \tag{2.38}$$

The constant C drops out completely.

2.9 Why the Newtonian Derivation Works

Near the end the question becomes: why should the Newtonian derivation agree with the relativistic one as well as it does?

Question from the audience. This all seems to depend on an assumption of an open universe.

[a] **Response.** At this stage the derivation is Newtonian, and Newtonian physics means flat space. The interpretation of k in terms of the geometry of space is postponed to the next lecture. There $k = 0$ will correspond to flat space, while the other signs will be tied to curved spatial geometry.

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[a]Lecture timestamp: 01:45:44.

Question from the audience. Sometime long ago, did the length of meter sticks depend on a?

[a] **Response.** Far enough back, matter would have been so crowded that separate meter sticks would not really exist. Before that they would be squeezed together. The lecture also reminds us that relativity does not allow truly rigid bodies anyway: if a body were perfectly rigid, a push on one side would send a signal to the other side instantaneously, faster than light.

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[a]Lecture timestamp: 01:47:53.

Question from the audience. Why do you think there is this coincidence of equations, Newtonian and relativistic?

[a] **Response.** Because the Newtonian treatment is being applied only to a sufficiently small region. As long as the region is small compared with the radius of curvature of the whole universe, and as long as local Hubble velocities there are well below the speed of light, Newtonian mechanics is a good approximation. The lecture compares this with studying the growth of grass on the Earth: one can understand the local process without using the curvature of the whole Earth.

[a]Lecture timestamp: 01:49:03.

This is the logic of the local patch. Newtonian mechanics assumes, first, that the curvature of space is not important for the question at hand, and second, that nothing locally is moving close to the speed of light. In cosmology that means choosing a region large enough to be homogeneous, but still small enough that curvature effects are negligible and Hubble velocities remain nonrelativistic.

With that in mind, the Newtonian calculation is not an accident. It is the local approximation to an equation that can also be read in relativistic form. If the curvature term is moved to the left, the Friedmann equation becomes

$$\left(\frac{\dot{a}}{a}\right)^2 + \frac{k}{a^2} = \frac{8\pi G}{3}\rho. \tag{2.39}$$

The left-hand side is geometric: it involves the scale factor and, next time, the curvature of space. The right-hand side is the energy density. That is exactly the kind of equation one expects from general relativity.

Next time the subject is curvature itself, Einstein's equations, and the relation between that relativistic derivation and the Newtonian one.

Chapter 3

Friedmann Expansion, Radiation, and Dark Matter

3.1 Review of the Basic Cosmological Equation

On scales larger than clusters of galaxies, the universe is taken to be homogeneous. Choose coordinates so that we sit at $x = 0$, and place a representative galaxy, or simply an imaginary particle moving with the Hubble flow, at $x = 1$. Its physical distance from us is then, by definition, the scale factor $a(t)$. A point at $x = 2$ is at distance $2a(t)$, and so on.

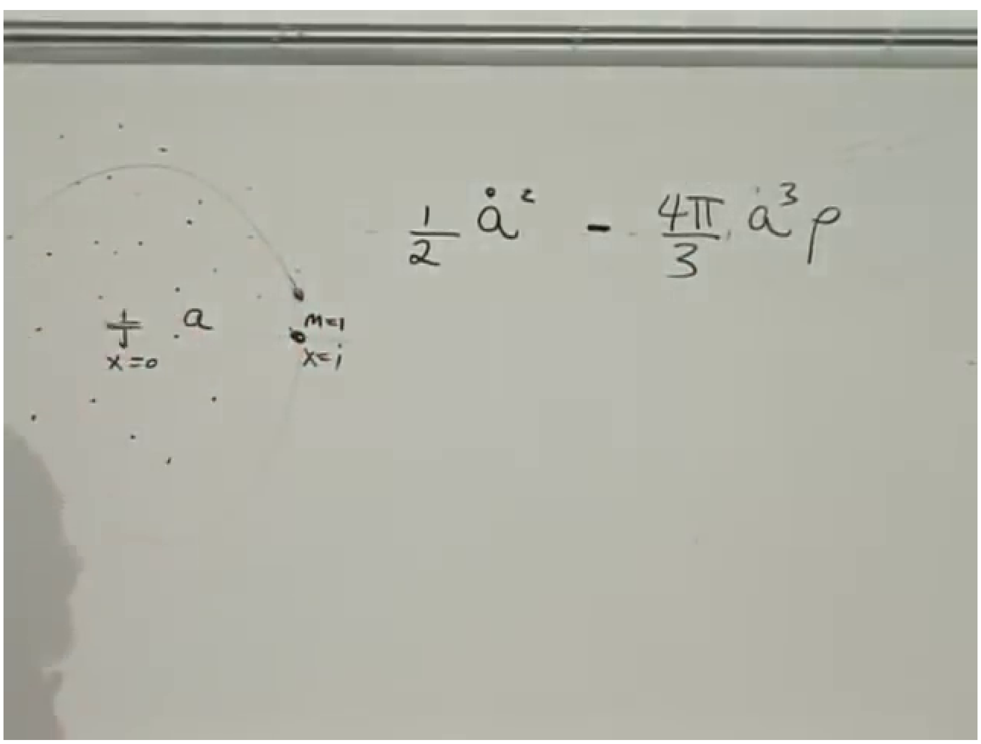

Figure 3.1: Homogeneous sphere centered at $x = 0$, a comoving galaxy at $x = 1$, and the Newtonian energy balance for $a(t)$.

Only the matter inside the sphere of radius $a(t)$ contributes to the gravitational pull on the particle at $x = 1$. Give that particle unit mass. Its kinetic energy is

$$\frac{1}{2}\dot{a}^2. \tag{3.1}$$

The mass enclosed by the sphere is

$$M_{\text{enc}} = \frac{4\pi}{3}a^3\rho, \tag{3.2}$$

so the gravitational potential energy is

$$-\frac{GM_{\text{enc}}}{a} = -\frac{4\pi G}{3}a^2\rho. \tag{3.3}$$

[1]

Energy conservation gives

$$\frac{1}{2}\dot{a}^2 - \frac{4\pi G}{3}a^2\rho = \text{const}. \tag{3.4}$$

The constant is then written as $-k$, with the usual warning that the standard normalization really fixes that notation only after multiplying the equation by 2. In the standard form,

$$\dot{a}^2 = \frac{8\pi G}{3}a^2\rho - k, \tag{3.5}$$

and dividing by a^2 gives

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho - \frac{k}{a^2}. \tag{3.6}$$

This is the Friedmann equation in the form used throughout the discussion.

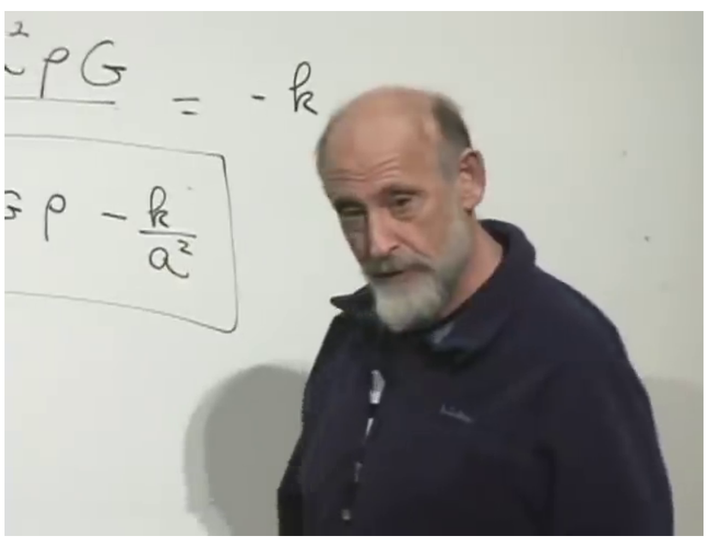

Figure 3.2: Curvature contribution $-k/a^2$ isolated in the Friedmann equation.

The curvature term is empirically very close to zero, close enough that at this level it is a very good approximation simply to neglect it. It does correspond to the curvature of space, but observationally it is too small here to determine even its sign.

[1] Editorial clarification: At lecture timestamp 00:05:30 the board clearly shows the kinetic term and the density-dependent potential term. The full coefficient written here follows from the spoken derivation.

The same equation also comes from general relativity. In schematic form, the time-time Einstein equation is

$$R_{00} - \frac{1}{2} g_{00} R \propto T_{00}. \tag{3.7}$$

[2] With an expanding homogeneous metric, the geometric side produces the combination of $\dot{a}/a$ and k/a^2, while the matter side produces the energy density. At this stage, the Newtonian argument and the Einstein equation are saying the same thing.

That also changes the meaning of ρ. From here on, it is better to call ρ the energy density. For slowly moving matter in the comoving frame, that is just the rest-energy density.

3.2 Matter Domination

Now assume that the energy density is carried by matter at rest in the comoving frame. Take a cube one coordinate unit on a side. Its physical side lengths are $a(t)$, so its physical volume is $a^3(t)$. If the mass-energy inside that comoving cube is M, and matter moves with the flow rather than crossing the boundary, then

$$\rho_m = \frac{M}{a^3}. \tag{3.8}$$

This is the matter-dominated case. In this idealization the particles are at rest in the box, so the temperature is taken to be zero and the pressure is taken to be zero. Pressure would require particles crossing the boundary of the box.

With $k \approx 0$, the equation becomes

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3} \frac{M}{a^3}. \tag{3.9}$$

Try a power law,

$$a(t) = Ct^p. \tag{3.10}$$

Then

$$\frac{\dot{a}}{a} = \frac{p}{t}, \tag{3.11}$$

so the left-hand side scales like t^{-2}, while the right-hand side scales like t^{-3p}. Matching powers gives

$$3p = 2, \qquad p = \frac{2}{3}. \tag{3.12}$$

Therefore

$$a(t) \propto t^{2/3}. \tag{3.13}$$

Keeping the coefficient gives

$$C^3 = 6\pi G M. \tag{3.14}$$

The point of the solution is the power law. A matter-dominated universe expands like the two-thirds power of time, not linearly. This law describes a long intermediate era, from roughly 10^5 years after the Big Bang to a few billion years later. The present universe is not literally matter-dominated, because dark energy now dominates, but ordinary matter and a large stretch of cosmological history are described this way.

[2] Editorial clarification: The discussion gives only the tensor structure here, not the exact overall numerical coefficient.

3.3 Radiation Domination

Earlier than that, the energy density was dominated by radiation. The reason is simple. The universe contains an enormous number of photons, roughly 10^9 photons for every proton, and when the universe is compressed those photons become more energetic.

A photon has no rest mass, but it does have energy. Its energy is

$$E_\gamma = h\nu = \frac{hc}{\lambda}. \tag{3.15}$$

The crucial assumption is that as the universe expands, photon wavelengths stretch with it:

$$\lambda \propto a. \tag{3.16}$$

A wave in a box gives the right picture. If the box grows, the wavelength of the mode grows with the box. The same idea is used here for the expanding universe.

Since $\lambda \propto a$, the energy of each photon scales as

$$E_\gamma \propto a^{-1}. \tag{3.17}$$

Take again a comoving unit box. Photons pass in and out, but on average the number inside stays fixed. Then the total photon energy in the box also scales like a^{-1}. Dividing by the physical volume a^3, the radiation energy density scales as

$$\rho_r \propto a^{-4}. \tag{3.18}$$

That extra factor of a^{-1} is the whole difference. Matter dilutes because the universe expands. Radiation dilutes for that reason and also because each photon is redshifted.

The Friedmann equation for a radiation-dominated universe is therefore

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\frac{M_r}{a^4}. \tag{3.19}$$

[3]

Use the same trial form $a(t) = Ct^p$. The left-hand side again scales like t^{-2}, while the right-hand side scales like t^{-4p}. Matching powers gives

$$4p = 2, \qquad p = \frac{1}{2}. \tag{3.20}$$

So

$$a(t) \propto t^{1/2}. \tag{3.21}$$

Keeping the coefficient gives

$$C^4 = \frac{32\pi G}{3} M_r. \tag{3.22}$$

[3] Editorial clarification: The spoken discussion reuses the symbol M for this new constant. The subscript r is added here only to avoid confusion with the matter case.

[a] **Question from the audience.** Why is there a G in there?

Response. Because what gravitates is energy density, not rest mass alone. For slowly moving matter the distinction is easy to miss, but photons force it on us: they have no rest mass and still gravitate.

[a]Lecture timestamp: 00:35:41.

A box full of radiation makes the point concrete. Push on a box filled with photons. Once the box accelerates, the wall moving into the photons is struck more strongly than the wall moving away from them, so one has to push harder against the radiation pressure. The box therefore has extra inertia, and by the equivalence principle that extra inertia also means extra gravitational mass. This is the box-of-photons argument used here to say that energy gravitates.

[a] **Question from the audience.** Is either of these equations solvable with an a that decreases with time?

Response. Yes. If the curvature term k/a^2 is restored, there are solutions that expand and then contract. That is not what our universe appears to be doing, but the equation allows it.

[a]Lecture timestamp: 00:38:11.

[a] **Question from the audience.** What happened to conservation of energy?

Response. If the k term is restored and the equation is multiplied back by a^2, the cosmological equation is precisely the conservation law. Matter or radiation on one side is only part of the bookkeeping. The expansion contributes the kinetic and gravitational part, and the total balance is conserved.

[a]Lecture timestamp: 00:38:44.

3.4 Matter and Radiation Together

Real cosmology contains both matter and radiation. Their densities scale differently:

$$\rho_m \propto a^{-3}, \qquad \rho_r \propto a^{-4}. \tag{3.23}$$

So at small a, radiation dominates. At large a, matter dominates. Running backward, the early universe must therefore have been radiation-dominated.

[4]

The matter curve falls like a^{-3}. The radiation curve falls like a^{-4}. Both diverge as $a \to 0$, but the radiation curve rises faster there, so the two curves cross at an equality scale factor. A rough estimate places this transition after roughly 10^4 years, perhaps a little more.

The same point can be expressed in the plot of $a(t)$ against t. Early on, the expansion behaves like $t^{1/2}$. Later it behaves like $t^{2/3}$. The change is not a jump; it is a continuous crossover through a relatively narrow range of time.

[4]Editorial clarification: At lecture timestamp 00:44:45 the board shows only the axes at the start of the comparison plot. The matter and radiation curves discussed immediately afterward are described in the text rather than redrawn here.

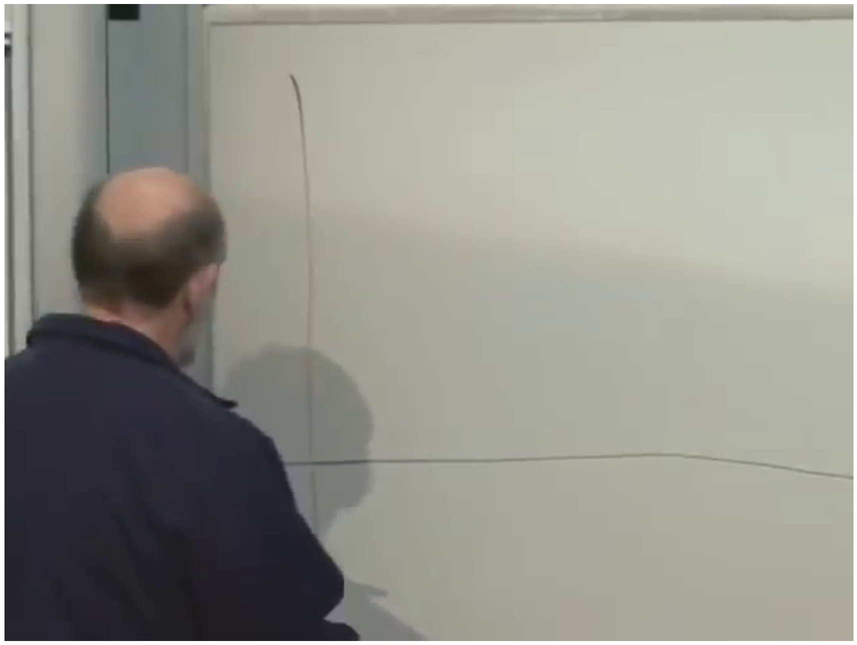

Figure 3.3: Axes for comparing matter and radiation energy density as functions of the scale factor.

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[a] **Question from the audience.** As the universe ages, does the power p go continuously from 1/2 to 2/3?
Response. Yes. It does not jump. The change is continuous, though it happens over a fairly narrow range of time.

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[a]Lecture timestamp: 00:44:04.

3.5 Temperature, Recombination, and Last Scattering

The next quantity is temperature. If the energy of a typical photon scales like a^{-1}, then so does the temperature of the radiation bath:

$$T \propto a^{-1}. \tag{3.24}$$

If a doubles, the temperature is cut in half.

Take the present photon temperature to be about 3 K, and the ionizing temperature to be about 3000 K, roughly the temperature of the surface of the sun. Then

$$\frac{a_{\text{today}}}{a_{\text{ion}}} \sim 10^3. \tag{3.25}$$

Using the late-time matter-dominated law $a \propto t^{2/3}$,

$$\left(\frac{t_{\text{today}}}{t_{\text{ion}}}\right)^{2/3} \sim 10^3, \tag{3.26}$$

so

$$\frac{t_{\text{today}}}{t_{\text{ion}}} \sim 1000^{3/2} \sim 3 \times 10^4. \tag{3.27}$$

Taking the age of the universe, at the level of this estimate, to be 10^{10} years gives

$$t_{\text{ion}} \sim 3 \times 10^5 \text{ years.} \tag{3.28}$$

This is the epoch when hydrogen and helium stop being fully ionized. Earlier than that, the universe is a hot plasma. Why is a plasma opaque? Because free charges respond strongly to electromagnetic waves. The electric field drives the charges back and forth, the medium behaves like a conductor, and conductors are opaque. That is the relevant analogy with the surface of the sun.

Once the universe cools enough for electrons to recombine with nuclei, ordinary neutral atoms form and the universe becomes much more transparent. The transition is not infinitely sharp. It is a fuzzy place, or a fuzzy time interval, but it is sharp enough for cosmology that it is called the surface of last scattering.

Looking outward means looking backward in time. Nearby galaxies show the recent past. More distant galaxies show earlier times. Far enough out, every line of sight reaches the epoch when the universe was still ionized and opaque. Electromagnetic radiation cannot show anything earlier than that.

Other messengers could in principle probe earlier times. Neutrinos are electrically neutral and would decouple earlier. Gravitons, if one could ever observe them, would come from still earlier epochs. But the same weakness of interaction that would let them pass through the early plasma also makes them extremely hard to detect.

If the universe was once as hot as the surface of the sun in every direction, why is the sky not blindingly bright? Because the last-scattering surface is receding, and the photons arriving from it have been redshifted by roughly the same factor of 10^3 that separates 3000 K from 3 K. What reaches us is microwave radiation, not visible sunlight.

[a] **Question from the audience.** So our telescopes cannot look earlier than about 300,000 years?

Response. Electromagnetic telescopes cannot. Before that time the universe was ionized and opaque. Other messengers, at least in principle, could probe earlier epochs.

[a]Lecture timestamp: 01:03:51.

[a] **Question from the audience.** When you talk about the temperature of a photon, you mean the temperature of a black body with that peak radiation. But that is not linear in wavelength, right?

Response. Right. The temperature is essentially the energy of a typical photon, so for a photon it scales inversely with wavelength. The T^4 law refers to the energy density, not the energy of a single photon. It combines the energy per photon with the number of photons per unit volume, and the number density scales like T^3.

[a]Lecture timestamp: 01:08:43.

Thus

$$n_\gamma \propto T^3, \qquad \rho_\gamma \propto T^4. \tag{3.29}$$

3.6 Dark Matter from Rotation Curves

Dark matter enters through galactic rotation curves. It was not inferred from the large-scale Friedmann equation, but from astronomy on galactic scales.

Start with the simplest Newtonian problem: a light body of mass m orbiting a central mass M at radius r. The gravitational force is

$$\frac{GMm}{r^2}, \tag{3.30}$$

and the centripetal acceleration is v^2/r. Newton's law gives

$$m\frac{v^2}{r} = \frac{GMm}{r^2}. \tag{3.31}$$

Cancel m, multiply by r, and obtain

$$v(r) = \sqrt{\frac{GM}{r}}. \tag{3.32}$$

So if the gravitating mass is concentrated near the center, the orbital speed falls like $r^{-1/2}$.

That is what one would expect if mass followed light. Most of the visible matter in a galaxy lies near the core, so outer stars would then behave much like planets in the outer solar system. Astronomers infer the orbital speeds by measuring Doppler-shift differences across a nearby galaxy, comparing the side moving toward us with the side moving away after subtracting the overall recessional motion of the galaxy as a whole. The measured orbital speeds stay roughly constant out to large radii.

So replace the central mass M by the enclosed mass $M(r)$:

$$\frac{v^2}{r} = \frac{GM(r)}{r^2}. \tag{3.33}$$

If $v(r)$ is approximately constant, then

$$M(r) \propto r. \tag{3.34}$$

In other words, the gravitating mass keeps increasing with radius even where the visible light does not.

Now assume, as the discussion does for simplicity, that the mass distribution is roughly spherical. Then

$$M(R) = \int_0^R 4\pi r^2 \rho(r)\, dr. \tag{3.35}$$

If $M(R) \propto R$, differentiating with respect to R gives

$$4\pi R^2 \rho(R) \propto 1, \tag{3.36}$$

and therefore

$$\rho(r) \propto \frac{1}{r^2}. \tag{3.37}$$

---

[a] **Question from the audience.** It looks as if the density should go like $1/r^2$. Is there a reason for that?

Response. From the observed flat rotation curve one infers $M(R) \propto R$, and differentiating then gives $\rho(r) \propto 1/r^2$. Beyond that, no simple first-principles explanation is claimed here; detailed simulations do much of the work.

---

[a] Lecture timestamp: 01:28:18.

The stable conclusion is that a galaxy contains much more gravitating mass than luminous matter alone would suggest, of order several times larger and roughly an order of magnitude larger in the estimate used here. The visible galaxy sits inside a much larger dark halo.

What keeps ordinary gas from staying in such a diffuse state? Dissipation. A gas cloud shrinks because collisions, friction-like processes, and radiation allow it to lose energy. Dark matter evidently does not radiate appreciably and is nearly collisionless. Gravity can pull it inward and convert potential energy into kinetic energy, but without efficient dissipation it does not collapse into the compact luminous structures made by ordinary matter. What keeps the halo extended is kinetic energy.

Several alternatives are then discussed cautiously. Neutrinos are said to be too fast to behave this way on galactic scales. A halo made from ordinary hydrogen would badly upset early-universe nucleosynthesis. Black holes made from ordinary matter are also mentioned as problematic, because they would imply too much ordinary matter in the early universe and would raise lensing issues; the discussion does not claim more than that on the spot.

The halo is not described as a thin shell or a torus. It is an approximately spherical, non-uniform ball, denser toward the center and falling gradually outward.

---

[a] **Question from the audience.** Are you talking about the Bullet Cluster?

Response. Yes. The X-ray gas shows the ordinary matter, while lensing shows the total gravitating mass. In the collision those two distributions separate. That is strong evidence for real dark matter rather than a mere modification of gravity tied only to visible matter.

---

[a] Lecture timestamp: 01:53:12.

That is the final observational point. Very little is known in detail about what dark matter is, but nothing here requires a radical new principle. The conservative possibility is simply that nature contains additional weakly interacting particles that have not yet been identified.

Chapter 4

Horizons, Equation of State, and Vacuum Energy

4.1 Preferred Frame, Superluminal Recession, and the Horizon

The cosmic microwave background does pick out a special frame. In the frame where that radiation looks isotropic, the nearby galaxies are also, on average, at rest. If we move through the background, photons in the forward direction are Doppler shifted to higher energy and photons in the backward direction to lower energy. The preferred frame is the one in which the microwave background looks isotropic, and it is also, on average, the frame in which the local galaxies are at rest.

[a] **Question from the audience.** Does the apparent wavelength of the cosmic radiation depend on the observer's motion, and does that imply a preferred frame?

Response. Yes. Motion through the microwave background produces a Doppler anisotropy, so there is a preferred cosmological frame: the one in which the microwave background looks isotropic. It is also, on average, the frame in which the local galaxies are at rest.

[a]Lecture timestamp: 00:00:25.

A second question concerns recession speeds. Hubble's law says that sufficiently distant galaxies separate according to

$$v = Hd. \tag{4.1}$$

For large enough d, this recession speed exceeds the speed of light. That does not contradict relativity, because the prohibition is local: one object cannot pass another right in front of us faster than light. The statement about distant galaxies is a statement about the geometry between widely separated places.

[a] **Question from the audience.** Can two galaxies have a relative velocity greater than c?

Response. Yes, if they are far enough apart. Hubble recession can exceed the speed of light without violating local relativity, and the place where the recession speed reaches c is a horizon.

[a]Lecture timestamp: 00:04:48.

Once recession reaches the speed of light, signals sent toward us no longer make net progress. The ray still moves locally at the limiting speed, but the distance it must overcome grows too quickly.

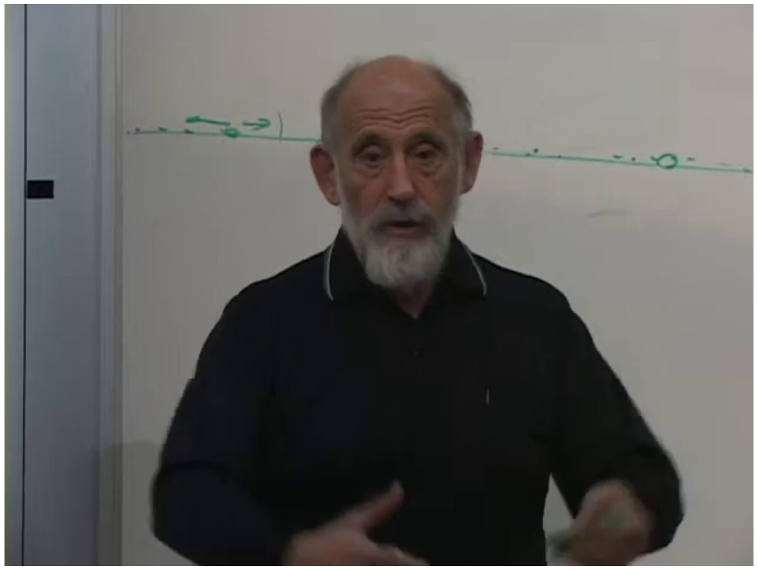

Figure 4.1: One-dimensional line used to picture an expanding metric.

[a] **Question from the audience.** Wouldn't a light ray sent in the reverse direction still travel toward us?

Response. Locally it does, but space grows faster than the ray can overcome. The ray moves at the limiting speed relative to nearby points, while the total distance back to us keeps increasing.

[a]Lecture timestamp: 00:07:03.

4.2 The Expanding Metric and the Line Analogy

A useful picture is one-dimensional. Imagine a line made of atoms. If we stretch an ordinary material line, eventually it breaks. Now modify the picture: every time neighboring atoms separate too much, we insert another one between them. The line keeps growing, but its local character does not thin out.

In this picture there is still a local limiting signal speed, just as sound on an ordinary line has a limiting speed. The rule is local: a disturbance cannot outrun that speed relative to nearby points. Nevertheless, if the line is long enough and new points are inserted rapidly enough, sufficiently distant parts of it recede so fast that a signal can never get back.

Special-relativistic velocity addition is therefore only part of the story. It applies locally, when observers pass one another at the same point. It is not the right tool for widely separated galaxies in an expanding geometry.

[a] **Question from the audience.** Does the ordinary relativistic law for adding velocities still hold in this situation?

Response. Locally, at a point, yes. If observers pass one another one after another, the usual special-relativistic rule applies. But it is not the right way to describe widely separated recession faster than c; there one has to go back to the geometry of expanding space.

[a]Lecture timestamp: 00:09:42.

The same picture gives a simple account of redshift. Draw a wave on the line. As the line expands, the wavelength expands with it. In an expanding geometry this is not merely a suggestive image. Wave equations in expanding spacetime really do produce stretched wavelengths.

---

[a] **Question from the audience.** Is the statement that a wave stretches with the universe theoretical, empirical, or both?

Response. It is a theoretical consequence of wave propagation in expanding spacetime, and it is also supported by astronomical observation.

---

[a]Lecture timestamp: 00:14:42.

This picture is not a rubber band. A rubber band thins out when stretched, and someone living in such a medium would see the local properties change. Here the local properties stay the same while the global distance scale grows.

4.3 Last Scattering and the Observable Universe

The horizon is the place where recession reaches the speed of light. The surface of last scattering lies slightly inside it. A signal emitted from the horizon would be stretched without bound; the last-scattering radiation has a huge but finite redshift.

---

[a] **Question from the audience.** What is the relationship between superluminal recession, the surface of last scattering, and the observable universe?

Response. The horizon is where recession reaches the speed of light. The surface of last scattering lies slightly inside it, because its redshift is enormous but finite. For practical purposes it marks the edge of the observable universe.

---

[a]Lecture timestamp: 00:17:53.

The redshift of the last-scattering radiation is roughly a factor of 10^3. That is already so large that the surface of last scattering is, for many purposes, very close to the horizon. The observable universe is the portion of the universe from which light can reach us.

At late times the expansion approaches the vacuum-dominated regime. Then the Hubble quantity tends toward a constant, the horizon approaches a fixed size, and both galaxies and microwave radiation are progressively lost from view.

---

[a] **Question from the audience.** Since H changes with time, is the horizon distance changing, and will the microwave background eventually disappear?

Response. At late times H tends to a constant, so the horizon approaches a fixed distance. The microwave background keeps stretching, cooling, and fading away. Distant galaxies also pass through the horizon and disappear from view.

---

[a]Lecture timestamp: 00:22:43.

Only structures that are already gravitationally bound remain visible. The solar system does not fly apart, and the galaxy does not fly apart, but the large-scale universe becomes an isolated island.

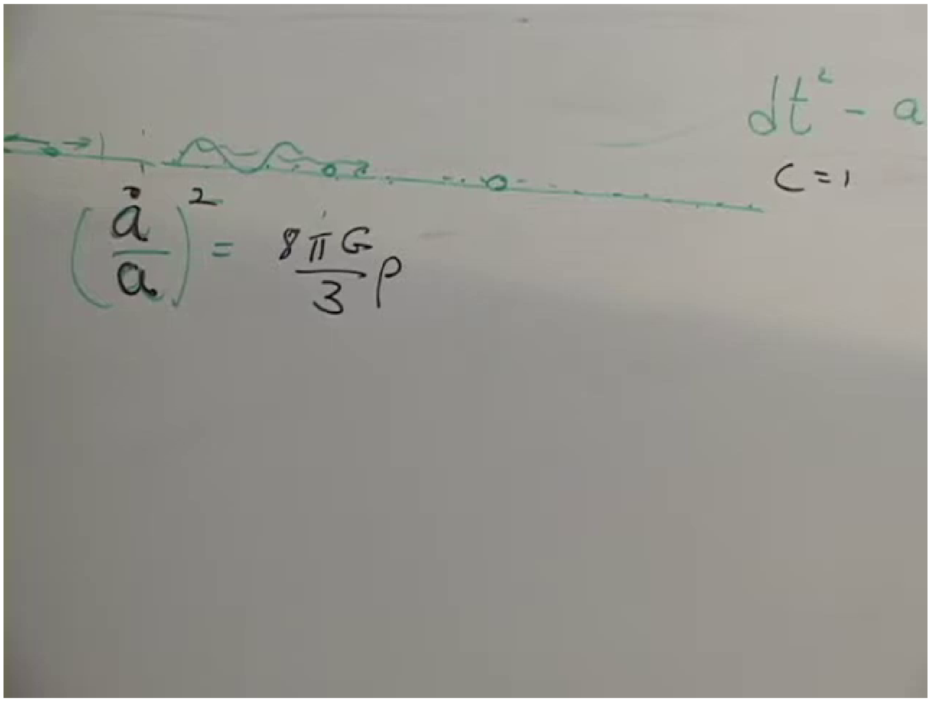

Figure 4.2: Basic expansion equation together with the one-dimensional line analogy.

4.4 The Basic Expansion Equation and the Three Components

Now the basic expansion equation is

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho, \qquad c = 1. \tag{4.2}$$

Here $a(t)$ is the scale factor, and ρ is the total energy density.

There are three dynamically important components.

For matter,

$$\rho = \frac{\text{const.}}{a^3}, \qquad a(t) \sim t^{2/3}. \tag{4.3}$$

This is the case of massive objects moving with the overall expansion and essentially at rest in the comoving frame.

For radiation,

$$\rho = \frac{\text{const.}}{a^4}, \qquad a(t) \sim t^{1/2}. \tag{4.4}$$

The extra power of a comes from the redshifting of each photon as the universe expands.

There is also

$$\rho_0 = \text{constant}, \tag{4.5}$$

the vacuum energy, also called dark energy or, up to conventional numerical factors, the cosmological constant. This component does not dilute at all.

The next step is to relate energy density to pressure.

4.5 Radiation Pressure and Cold Matter

Start with radiation. Consider a one-dimensional box, just a line segment of length L, containing photons bouncing back and forth. In one dimension, force and pressure are effectively the same thing.

Take a photon of momentum magnitude p. When it reflects from a wall, its momentum changes by

$$\Delta p = 2p. \tag{4.6}$$

The time between successive collisions with the same wall is

$$\Delta t = \frac{2L}{c}. \tag{4.7}$$

Therefore the average force on the wall is

$$F = \frac{\Delta p}{\Delta t} = \frac{pc}{L}. \tag{4.8}$$

For a photon,

$$E = pc, \tag{4.9}$$

so

$$F = \frac{E}{L}. \tag{4.10}$$

The energy divided by the length is the one-dimensional energy density. In this one-dimensional case,

$$P = \rho. \tag{4.11}$$

With many photons the same relation survives after averaging. The total force is just the total energy per unit length.

Now go to three dimensions. If all photons moved along one axis only, they would exert pressure on one pair of faces and none on the others. That is deliberately wrong, but it shows immediately what has to be corrected. In an isotropic bath, momentum is distributed over all directions. On average, one third of the energy density contributes to each pressure component. The result is

$$P = \frac{\rho}{3}. \tag{4.12}$$

For cold matter the answer is different. Take a comoving box, a small region moving with the flow of the universe. The particles inside carry energy, but if they are at rest relative to the box they do not bang into its walls. The pressure is therefore

$$P = 0. \tag{4.13}$$

The board summary is

$$\text{matter:} \qquad a(t) \sim t^{2/3}, \qquad P = 0, \tag{4.14}$$

$$\text{radiation:} \qquad a(t) \sim t^{1/2}, \qquad P = \frac{\rho}{3}. \tag{4.15}$$

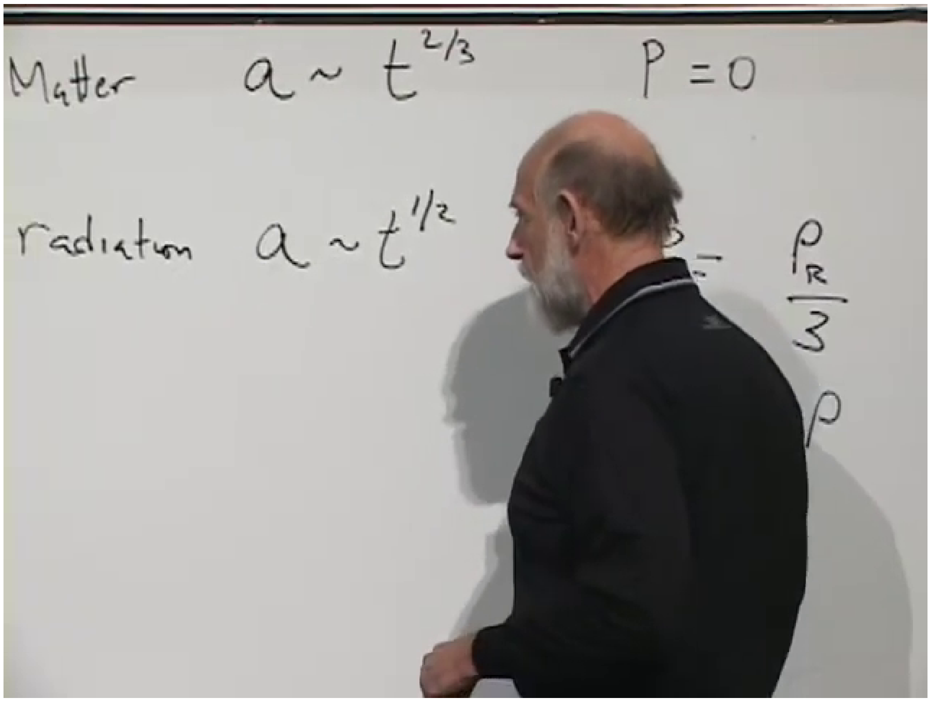

Figure 4.3: Matter and radiation: $a \sim t^{2/3}$, $P = 0$, and $a \sim t^{1/2}$, $P = \rho/3$.

---

[a] **Question from the audience.** In the expanding universe, aren't the particles actually moving apart?

Response. Yes, but the pressure argument is applied to an infinitesimal comoving box. On that local scale the thermodynamic reasoning still works.

---

[a]Lecture timestamp: 00:54:27.

4.6 The Equation of State

Pressure is taken to be proportional to energy density,

$$P = w\rho. \tag{4.16}$$

On the board the same parameter is written as ω. For the two basic cases,

$$w = 0 \quad \text{for cold matter,} \qquad w = \frac{1}{3} \quad \text{for radiation.} \tag{4.17}$$

Intermediate cases are possible. Values between 0 and 1/3 describe material that is neither completely cold nor fully relativistic. Warm dark matter is the example mentioned here.

The next question is whether w controls the dependence of ρ on a. The answer comes from a thermodynamic identity.

Take a box of volume V. If the contents exert pressure P, then when the box expands by dV the work done on the walls is

$$dE = -P\,dV. \tag{4.18}$$

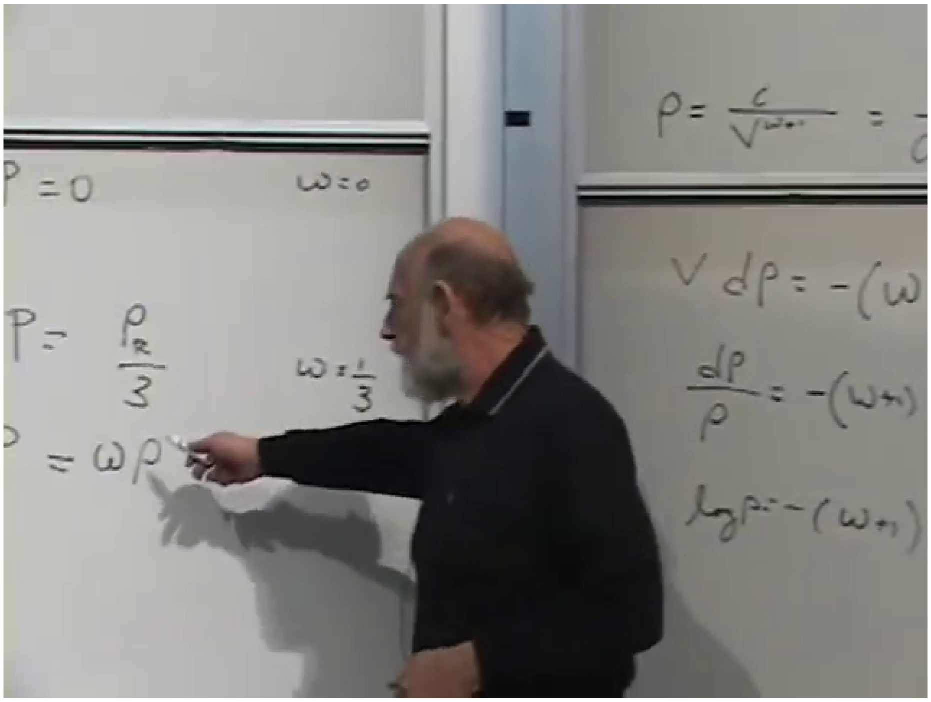

Figure 4.4: Equation of state $P = \omega\rho$ together with the density-scaling argument.

Now write the energy as

$$E = \rho V. \tag{4.19}$$

Its differential is

$$dE = \rho\, dV + V\, d\rho. \tag{4.20}$$

Equating the two expressions gives

$$\rho\, dV + V\, d\rho = -P\, dV, \tag{4.21}$$

or

$$V\, d\rho = -(\rho + P)\, dV. \tag{4.22}$$

Using $P = w\rho$,

$$V\, d\rho = -(1 + w)\rho\, dV. \tag{4.23}$$

Divide by ρV:

$$\frac{d\rho}{\rho} = -(1 + w)\frac{dV}{V}. \tag{4.24}$$

Integrating,

$$\rho \propto V^{-(1+w)}. \tag{4.25}$$

For a comoving cosmological box,

$$V \propto a^3, \tag{4.26}$$

and therefore

$$\rho \propto a^{-3(1+w)}. \tag{4.27}$$

The earlier cases are recovered at once:

$$w = 0 \Longrightarrow \rho \propto a^{-3}, \tag{4.28}$$

$$w = \frac{1}{3} \Longrightarrow \rho \propto a^{-4}. \tag{4.29}$$

This is why w matters. Once $\rho(a)$ is known, it can be put back into the Friedmann equation. One then tries a power law $a \sim t^{\alpha}$ and determines how the universe expands for that equation of state.

4.7 Vacuum Energy and Negative Pressure

One special case behaves differently:

$$w = -1. \tag{4.30}$$

Then

$$1 + w = 0, \tag{4.31}$$

so the exponent vanishes and

$$\rho = \rho_0 = \text{constant}. \tag{4.32}$$

The corresponding pressure is

$$P = -\rho. \tag{4.33}$$

Negative pressure is not mysterious in itself. Positive pressure pushes outward on the walls of a box. Negative pressure pulls inward. It is tension. A stretched rubber band gives the everyday example: it stores positive energy, but it pulls its ends together.

Substitute the constant density ρ_0 into the Friedmann equation:

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho_0. \tag{4.34}$$

The right-hand side is just a number, so

$$\frac{\dot{a}}{a} = \sqrt{\frac{8\pi G}{3}\rho_0}. \tag{4.35}$$

This is the special case in which the Hubble quantity really is constant. The equation can be written as

$$\dot{a} = \sqrt{\frac{8\pi G}{3}\rho_0}\, a, \tag{4.36}$$

and the solution is

$$a(t) \propto \exp\left(\sqrt{\frac{8\pi G}{3}\rho_0}\, t\right). \tag{4.37}$$

This is exponential expansion. The relevant timescale is the e-folding time, not a doubling time, and the rough estimate used here puts it on the order of the age of the universe. Vacuum energy and cosmological constant are the same physical ingredient apart from historical numerical conventions.

[a] **Question from the audience.** If vacuum energy drives exponential expansion, won't everything eventually be ripped apart?

Response. Not necessarily. Bound systems can resist it. A galaxy, a solar system, or an atom need not be torn apart, whereas sufficiently large and dilute structures do participate in the expansion and eventually separate.

[a]Lecture timestamp: 01:27:16.

[a] **Question from the audience.** Is there anything interesting in the negative sign when one takes the square root of the Friedmann equation?

Response. Yes. The square root gives two branches. One is the expanding solution, and the other is the time-reversed contracting solution. The equations themselves admit both.

[a]Lecture timestamp: 01:31:43.

4.8 Three Eras and the Observed Expansion History

All three components are present, and they scale differently:

$$\begin{aligned} \rho_{\text{radiation}} &\propto a^{-4}, \\ \rho_{\text{matter}} &\propto a^{-3}, \\ \rho_0 &= \text{constant}. \end{aligned} \tag{4.38}$$

That immediately implies three eras.

At very small a, radiation dominates. Later, matter dominates. At sufficiently large a, vacuum energy dominates. The corresponding expansion laws are

$$a(t) \sim t^{1/2}, \qquad a(t) \sim t^{2/3}, \qquad a(t) \sim e^{Ht}. \tag{4.39}$$

The rough times quoted here are these: radiation gives way to matter after about 10^4 years. The universe becomes transparent later, after about 10^5 years. The transition from matter domination to accelerated expansion occurs much later, after several billion years, roughly halfway through cosmic history in the order-of-magnitude estimate being used.

How is this history inferred? We cannot see the future, but we can see the past. Looking to greater distance means looking to earlier time. By observing distant galaxies, measuring how their motion correlates with distance, and reconstructing the Hubble quantity as a function of lookback time, one reconstructs the time history of the scale factor. The radiation-dominated era is too early to be seen directly in ordinary galaxy astronomy, but the matter-dominated era and the transition to vacuum domination are visible in the reconstructed curve.

4.9 Last Scattering in the Far Future

The future microwave sky is not the same story as galaxies disappearing through the horizon. The radiation we see at later times comes from farther matter, so the effect is seeing deeper and cooler rather than watching the same atoms simply recede away.

[a] **Question from the audience.** If the surface of last scattering fades into the horizon, is that the same thing as galaxies moving beyond the horizon?

Response. No. The later radiation does not come from the same atoms as the radiation we see now. It comes from atoms that were farther away, so it is like seeing deeper and deeper into the universe, while the radiation looks cooler and cooler.

[a] Lecture timestamp: 01:48:07.

4.10 Curvature, the Big Crunch Picture, and the Late-Time Horizon

Up to this point the curvature term has been suppressed. Restoring it gives

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho - \frac{K}{a^2}. \tag{4.40}$$

Numerically this term seems small, but conceptually it matters. In the older Newtonian language it distinguishes the cases of positive, zero, and negative total energy. In the relativistic language it distinguishes the spatial geometries.

Before vacuum energy was taken seriously, one often imagined a matter-dominated universe in which the expansion would eventually reverse. The result was the old Big-Crunch picture: expansion from the Big Bang, then recontraction. The mathematics is still valid, but it no longer appears to describe our universe.

[a] **Question from the audience.** Is this vacuum energy related to quantum vacuum energy?

Response. Yes, or at least it is believed to be.

[a] Lecture timestamp: 01:53:51.

The rough composition today is

$$\frac{\rho_0}{\rho_{\text{total}}} \sim 0.7, \qquad \frac{\rho_{\text{matter}}}{\rho_{\text{total}}} \sim 0.3. \tag{4.41}$$

Of that matter component, luminous matter is about 0.05 and dark matter about 0.25 of the total. Radiation is negligible today. Dark and luminous matter scale in the same way, so their ratio stays fixed even while the relative abundance of matter, radiation, and vacuum changes from era to era.

In the vacuum-dominated limit,

$$H = \sqrt{\frac{8\pi G}{3}\rho_0}. \tag{4.42}$$

Then Hubble's law becomes

$$v = Hd. \tag{4.43}$$

The horizon is defined by $v = 1$, so its distance is

$$d_H = \frac{1}{H} = \frac{1}{\sqrt{8\pi G\rho_0/3}}. \tag{4.44}$$

The rough number given here is about 10 to 12 billion light years.

The important point is the late-time behavior. Once the expansion is exponential, the horizon distance becomes fixed. Galaxies continue to pass through it and disappear by becoming more and more redshifted. The microwave background is redshifted away as well. In the remote future, only the nearby gravitationally bound structures remain visible, and the universe looks like an isolated island.

4.11 Geometry from the Last-Scattering Surface

Finally the geometry can be tested from the last-scattering surface. Suppose the distance to that surface is known. Suppose also that one can identify a lump there whose physical size is known well enough. Then one can compare that size and distance with the observed angular size.

In flat space the angle is the Euclidean one. In positively curved space it would appear larger; in negatively curved space, smaller. The analogy is ordinary surveying: take an object of known size at known distance and compare the measured angle with Euclidean geometry.

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[a] **Question from the audience.** What is an example of an object out there whose size we know?

Response. The largest lumps in the microwave background. We think we know their size from how long they took to evolve and from the speed of sound.

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[a]Lecture timestamp: 02:08:09.

The relevant objects here are the largest lumps on the microwave background. Their observed angular size comes out consistent with Euclidean geometry to within a few percent. On that basis, space appears flat.

4.12 Measured w and a Final Question

The observational importance of w is now clear. If astronomers report that w is very close to -1, they are saying that the vacuum-energy component is very nearly constant. That is exactly the point made here.

That does not mean general relativity forces w to be exactly -1. The point here is empirical. The lecture briefly notes that more radical possibilities are not excluded in principle, but that the nearly constant vacuum-energy case is the one for which there is good reason to believe.

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[a] **Question from the audience.** In the density calculation for matter and for radiation, did we assume that the energy was constant and then adjust the scale factor to figure out the formula? And if that is true for the first two cases, what happens in the vacuum-energy case, where the density stays constant?

Response. No. I did not assume it was constant. We used that the change in energy is $-P\,dV$. For an imaginary comoving box moving along with the expansion, work is done on the walls, and that accounts for the change in energy. The vacuum-energy case raises the question in a different way, and the closing remark here is that energy-conservation language in cosmology does not say very much, because the total energy of the universe is zero.

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[a]Lecture timestamp: 02:11:04.

Chapter 5

Spheres, Hyperbolic Space, and the Meaning of k

Let us do a little mathematics, some simple mathematics. Spheres in any number of dimensions are special geometries: they are uniform, and here uniform means constant curvature. Positive curvature comes first, then uniform negative curvature, and after that the connection with Einstein's equations.

A spherical universe means that *space* is spherical, not spacetime. There is also the other possibility, uniform negative curvature. Putting these cases side by side is what makes the role of general relativity clear.

5.1 Spheres and Uniform Geometry

A one-sphere is just a circle. The simplest way to describe it is to embed it in a two-dimensional auxiliary space and write

$$x^2 + y^2 = R^2. \tag{5.1}$$

The unit one-sphere is

$$x^2 + y^2 = 1. \tag{5.2}$$

A two-sphere is described the same way, now using three auxiliary coordinates:

$$x^2 + y^2 + z^2 = R^2, \tag{5.3}$$

with unit case

$$x^2 + y^2 + z^2 = 1. \tag{5.4}$$

These coordinates are only a crutch. The geometry of interest is not the surrounding x, y, z space. It is the surface itself. A creature living on a one-sphere lives on a circle; it has no access to an inside or an outside. The same idea applies to a two-sphere. The filling of the sphere and the outside of the sphere belong only to the auxiliary picture.

The same construction gives a three-sphere. One introduces one more coordinate, say w, and writes

$$x^2 + y^2 + z^2 + w^2 = R^2, \tag{5.5}$$

with unit case

$$x^2 + y^2 + z^2 + w^2 = 1. \tag{5.6}$$

A three-sphere is harder to visualize, because the crutch would now be four-dimensional, but the mathematics is not much harder.

The standard notation is Ω_n for the n-sphere. Thus Ω_1 is the one-sphere, Ω_2 the two-sphere, and Ω_3 the three-sphere.

5.2 Metrics on the Sphere

The more useful description is intrinsic. One describes the sphere by coordinates that live on the sphere itself.

For the unit one-sphere one coordinate ϕ is enough, with $0 \le \phi \le 2\pi$. Its metric is

$$ds^2 = d\phi^2. \tag{5.7}$$

This is also written

$$d\Omega_1^2 = d\phi^2. \tag{5.8}$$

The unit two-sphere is described as a sequence of one-spheres. One angle ϕ still goes around each circle, but a second angle θ runs from one end of the sphere to the other. That second angle goes from 0 to π. Motion in the θ-direction contributes $d\theta^2$. Motion around the ϕ-direction takes place on a circle whose radius is $\sin\theta$. Thus

$$d\Omega_2^2 = d\theta^2 + \sin^2\theta\, d\phi^2 \tag{5.9}$$

or

$$d\Omega_2^2 = d\theta^2 + \sin^2\theta\, d\Omega_1^2. \tag{5.10}$$

The three-sphere is obtained the same way, now as a sequence of two-spheres. Introduce another coordinate α. Then

$$d\Omega_3^2 = d\alpha^2 + \sin^2\alpha\, d\Omega_2^2. \tag{5.11}$$

Expanded out, this is

$$d\Omega_3^2 = d\alpha^2 + \sin^2\alpha \left(d\theta^2 + \sin^2\theta\, d\phi^2\right). \tag{5.12}$$

Here α runs from one zero-size end to the other, so $0 \le \alpha \le \pi$.[1]

The general recursive form is

$$d\Omega_n^2 = d\alpha^2 + \sin^2\alpha\, d\Omega_{n-1}^2. \tag{5.13}$$

Once this point of view is adopted, the embedding picture can be forgotten. A one-sphere is described by one coordinate, a two-sphere by two, a three-sphere by three, and so on. The geometry stands on its own.

This geometry is special because it is homogeneous. One could deform the metric by replacing $\sin^2\theta$ with some other function that still vanishes at both ends. One would then get some deformed closed

[1] Editorial clarification: At 00:17:24 the audio briefly sounds like "zero to two pi," but the surrounding construction and the explicit $\sin^2\alpha$ factor require the new coordinate to run from one zero-size end of the family of lower-dimensional spheres to the other.

surface, for example an ellipsoid. But it would no longer be uniform. The sphere is homogeneous because every point is the same as every other point.

If one wants a sphere of radius A instead of unit radius, one multiplies the unit metric by A^2:

$$ds^2 = A^2\, d\Omega_3^2. \tag{5.14}$$

The square is necessary because the metric measures squared distance.

A brief aside concerns the zero-sphere. Extrapolating one step backward gives

$$x^2 = 1, \tag{5.15}$$

whose solutions are the two points $x = \pm 1$. So the unit zero-sphere is just two points.

5.3 A Spherical FRW Universe

Now the interest is not in space alone, but in spacetime. The relevant picture is a three-sphere whose radius changes with time.

One warning matters throughout: time is different. One should not think of the expanding universe as a higher-dimensional sphere with time playing the role of one more embedding coordinate.

With that in mind, the spacetime interval is

$$d\tau^2 = dt^2 - a(t)^2\, d\Omega_3^2. \tag{5.16}$$

This is the FRW metric for a spatially spherical universe. The only new ingredient compared with the static sphere is that the radius has become a time-dependent function $a(t)$.

It is convenient to choose coordinates so that we are at one pole. Nothing is special about that point; it is just a convention. Then α can be thought of as a radial coordinate away from us on the unit sphere. It is not a radial distance in kilometers. It is a dimensionless angular measure on the unit sphere, but it plays the role of radial distance in the geometry.

Apart from local lumps such as galaxies and stars, this is the whole geometry of a spatially spherical FRW universe. Its spatial curvature is positive, and the sign convention is written

$$k = +1. \tag{5.17}$$

Nothing itself is "plus one." It simply means positive spatial curvature.

5.4 Looking Out from a Sphere

How does such a sphere look when one looks out from one point on it? This is where the stereographic projection enters.

Take a two-sphere tangent to an infinite plane at the south pole. Pick a point on the sphere. Draw the straight line from the north pole through that point and continue it until it hits the plane. In this way each point on the sphere is mapped to a point on the plane. The south pole lands at the origin of the plane. The north pole is the exceptional point: as one approaches it from different directions, its image runs off to infinity in a direction-sensitive way. One therefore treats it as the point at infinity.

If equal "continents" are placed at different positions on the sphere, their images on the plane become larger as one moves toward the north pole. So an observer placed at the south pole would see distant objects subtend larger angles than they would in flat space. In positive curvature, a known object looks too big relative to Euclidean geometry.

That is the basic observational point. Curvature is inferred from the relation between distance, size, and angular size. Galaxies themselves are not good standard rulers for very deep distances, but the logic is exactly that.

[a] **Question from the audience.** Do you also have to know the distance, not just the angular size?

Response. Yes. The angle by itself is not enough. You need the angle and the distance.

[a]Lecture timestamp: 00:45:54.

[a] **Question from the audience.** Do circles on the surface of the sphere project as circles?

Response. Yes. Circles project to circles. That is one of the special features of stereographic projection.

[a]Lecture timestamp: 00:46:18.

[a] **Question from the audience.** Would a continent keep the same shape under the projection?

Response. There are two different statements. One is that circles project to circles. The other is the weaker statement that sufficiently small shapes preserve their local shape because angles are preserved. The mapping is conformal: it preserves angles, though not relative sizes.

[a]Lecture timestamp: 00:47:02.

5.5 Uniform Negative Curvature

The uniformly negatively curved case is introduced through its metric.

The negatively curved analog of the sphere is hyperbolic space. In the same recursive spirit as before, its metric is obtained by replacing $\sin\alpha$ with $\sinh\alpha$:

$$d\mathcal{H}_n^2 = d\alpha^2 + \sinh^2\alpha\, d\Omega_{n-1}^2. \tag{5.18}$$

The lower-dimensional angular piece remains the unit sphere metric $d\Omega_{n-1}^2$.

[a] **Question from the audience.** In the metric, is that lower-dimensional differential still the spherical one?

Response. Yes. That lower-dimensional piece is the ordinary unit-sphere metric. The change is in the $\sinh^2\alpha$ factor in front of it.

[a]Lecture timestamp: 00:55:51.

The hyperbolic sine is

$$\sinh\alpha = \frac{e^{\alpha} - e^{-\alpha}}{2}. \tag{5.19}$$

The minus sign matters; in the lecture it is corrected immediately by a student.

There is also one crucial difference in the range of the coordinate. For the sphere, the new coordinate runs from one end to the other and stops at π. Here,

$$0 \le \alpha < \infty. \tag{5.20}$$

The ordinary sine rises and then comes back to zero. The hyperbolic sine begins the same way near $\alpha = 0$, but then it grows exponentially.

That means the lower-dimensional spheres around us grow bigger and bigger as we move outward. In this geometry the sky at increasing radius does not first grow and then shrink. It keeps growing.

In three spatial dimensions the corresponding FRW form is

$$d\tau^2 = dt^2 - a(t)^2\, d\mathcal{H}_3^2, \tag{5.21}$$

and the sign label is

$$k = -1. \tag{5.22}$$

5.6 The Disk Picture and What It Means

Negative curvature becomes intuitive in the disk picture made famous by Escher, especially *Limit Circle Number Four*. The analogy with stereographic projection is useful, but the behavior is opposite.

For the sphere, the projection looks infinite while the real geometry is finite. For the negatively curved case, the disk picture looks finite while the real geometry is infinite. Equal "continents" or "galaxies" appear smaller and smaller as one moves toward the boundary, but geometrically they are all the same size. If distance is measured by counting how many equal pieces one crosses, then the boundary is infinitely far away.

Thus a known object in negative curvature looks too small relative to flat space. It subtends too small an angle.

The geometry is still homogeneous. Although the picture makes the outer pieces look tiny, any observer moved to another location would see the same intrinsic geometry.

[a] **Question from the audience.** Would an observer in any one of those circles see a similar pattern?

Response. Yes, exactly the same thing. The space is homogeneous.

[a]Lecture timestamp: 01:06:42.

[a] **Question from the audience.** In the spherical case distant things seem too big, while here they seem too small?

Response. Yes. In positive curvature a known object subtends too large an angle relative to flat space. In negative curvature it subtends too small an angle.

[a]Lecture timestamp: 01:07:03.

[a] **Question from the audience.** Is the relevance of the stereographic-style projection that an astronomer can reconstruct it from observations?

Response. Yes. The astronomer is effectively a surveyor who surveys triangles in the sky. For the immediate geometric point one can set aside the time dependence and focus on angles and distances.

[a]Lecture timestamp: 01:08:09.

There is also a useful local picture. A small patch of negative curvature can be drawn as a saddle surface in ordinary three-dimensional space. But only small patches can be drawn that way; the whole hyperbolic plane cannot be embedded globally as a nice ordinary surface in three-dimensional Euclidean space.

5.7 The Flat Case

Between the positively curved and negatively curved cases sits flat space. One may think of it as a limit, but it is enough simply to think of ordinary flat Euclidean space.

Its FRW form is

$$d\tau^2 = dt^2 - a(t)^2 \left(dx^2 + dy^2 + dz^2\right), \tag{5.23}$$

with

$$k = 0. \tag{5.24}$$

These are the three homogeneous FRW geometries:

$$k = +1 : \text{positive spatial curvature,} \tag{5.25}$$

$$k = 0 : \text{flat space,} \tag{5.26}$$

$$k = -1 : \text{negative spatial curvature.} \tag{5.27}$$

[a] **Question from the audience.** If one projects the geometry onto a plane and follows a signal, would the projected speed of light vary from place to place?

Response. Yes, on the projection it would. On the spherical projection it would appear to go faster and faster farther out; on the negative-curvature disk it would appear to go slower and slower near the edge. That variation belongs to the picture, not to the underlying physics.

[a]Lecture timestamp: 01:15:07.

[a] **Question from the audience.** Has one actually used looking at far-away objects in this way to determine flatness?

Response. Yes, but it is really the angular information that is used. One does not determine flatness by trying to measure how fast light appears to move on the projection.

[a]Lecture timestamp: 01:17:05.

5.8 Back to the Cosmological Equations

After the break comes the connection between the cosmological equations already studied and Einstein's equations. At this point the board still carries the flat spatial metric fragment.

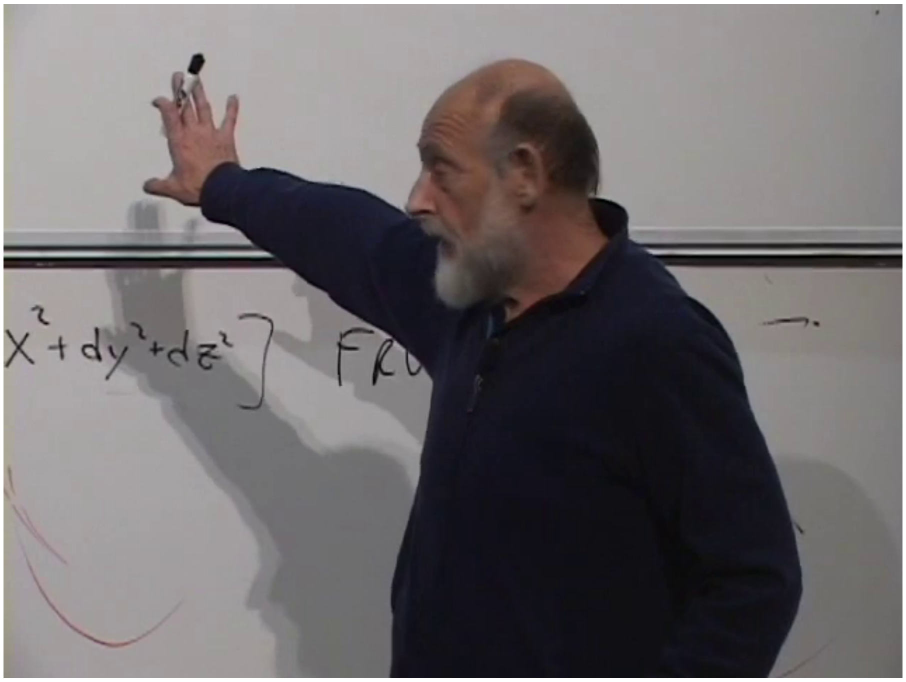

Figure 5.1: Flat FRW metric fragment.
Lecture frame: 01:18:18.

A very brief Newtonian review comes first. Put us at the center. Look at a galaxy moving away according to the Hubble law. By Newton's theorem for a spherically symmetric distribution, only the mass inside the sphere whose radius matches the galaxy's distance contributes to the force on it. Then there are three possibilities: the galaxy moves above the escape velocity, below the escape velocity, or exactly at the escape velocity.

What is recalled next is that for ordinary nonrelativistic matter the density goes like the mass divided by a^3, for radiation it goes like $1/a^4$, and for vacuum energy it does not change as the sphere changes.

In the earlier Newtonian treatment, the cosmological equation had a density term and, if one did not sit exactly at the escape-velocity case, an additional constant term proportional to $1/a^2$. The earlier discussion mostly set that term to zero. Empirically it is small, but conceptually there is no reason to leave it out.

5.9 Einstein's Equations and the 00 Component

The next step is Einstein's field equations. On the left-hand side stands the geometry, built from the spacetime curvature. On the right-hand side stands the energy-momentum tensor, the source of the gravitational field. In the lecture's normalization one writes

$$R^{\mu\nu} - \frac{1}{2} g^{\mu\nu} R = 4\pi G\, T^{\mu\nu}. \tag{5.28}$$

[2]

Not all ten independent components are needed here. By this stage so much has already been assumed — homogeneity, the FRW form, and a single time-dependent scale factor — that there is only one arbitrary function left, namely $a(t)$. The natural component to choose is the time-time component, because T^{00} is the energy density ρ.

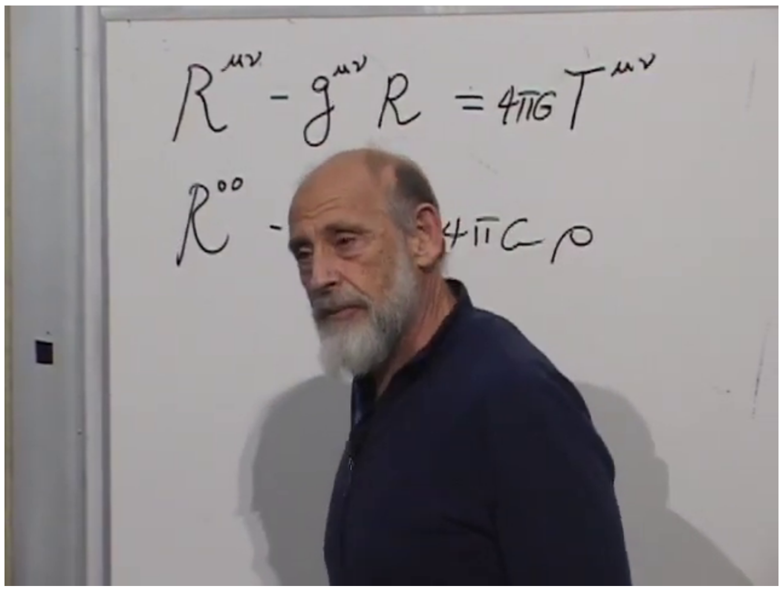

Figure 5.2: Einstein field equation and its 00 component.

Lecture frame: 01:30:54.

So the focus is

$$R^{00} - \frac{1}{2} g^{00} R = 4\pi G\, \rho. \tag{5.29}$$

The right-hand side is the energy density. The left-hand side is geometry.

[a] **Question from the audience.** What about the other components of the energy-momentum tensor?

Response. They do not vanish in general. The space-space components can represent pressure. In the matter-dominated case the pressure is zero, in the radiation-dominated case it is one-third of the energy density, and for vacuum energy one writes $P = w\rho$ with $w = -1$.

[a] Lecture timestamp: 01:39:09.

[a] **Question from the audience.** So if one had only matter, would the other components of $T^{\mu\nu}$ vanish?

Response. For pressureless matter, yes, the pressure components vanish.

[a] Lecture timestamp: 01:40:24.

[2] Editorial clarification: In the visible board line the factor $\frac{1}{2}$ is not legible, but around 01:33:36 the spoken correction restores it. The equation here keeps that correction while preserving the lecture's $4\pi G$ normalization.

5.10 What the Einstein Calculation Produces

The full curvature calculation is not carried out on the board. What matters is what appears when an FRW metric is substituted into the Einstein tensor.

One kind of term comes just from the fact that the scale factor depends on time. Even if space were flat, a time-dependent scale factor would still give spacetime curvature. That contribution is proportional to

$$\left(\frac{\dot{a}}{a}\right)^2. \tag{5.30}$$

The second kind of term comes from the spatial curvature itself. Even if the scale factor were frozen in time, curved space would still contribute to the spacetime curvature. For a simple uniformly curved geometry, curvature goes like one over the radius of curvature squared, so it contributes a term proportional to

$$\frac{k}{a^2}, \tag{5.31}$$

with a positive sign for the sphere, a negative sign for the hyperbolic case, and zero for flat space.

[a] **Question from the audience.** Should there also be a factor such as a radius of curvature?

Response. That is already the $1/a^2$ factor. For these simple uniformly curved geometries, curvature is just the inverse radius of curvature squared.

[a]Lecture timestamp: 01:36:53.

The coefficient bookkeeping is done on the fly and is partly checked against the earlier Newtonian answer. The endpoint is[3]

$$\left(\frac{\dot{a}}{a}\right)^2 + \frac{k}{a^2} = \frac{8\pi G}{3}\rho, \tag{5.32}$$

or equivalently

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho - \frac{k}{a^2}. \tag{5.33}$$

This is the form that links the three FRW geometries to the earlier cosmological dynamics.

[a] **Question from the audience.** Could k take any value if one redefined a?

Response. In the flat case there is no unique natural scale for a, so one can rescale it, but k remains zero. In the spherical and negatively curved cases there is a natural curvature scale. If one takes $a(t)$ to be that natural radius of curvature, then k is just $+1$ or -1.

[a]Lecture timestamp: 01:37:52.

[a] **Question from the audience.** Is k equal to $+1$, -1, or 0 in that final equation?

Response. Yes. It is $+1$ for the sphere, -1 for the hyperbolic case, and 0 for flat space.

[a]Lecture timestamp: 01:41:00.

[3]This displayed Friedmann form consolidates the spoken, self-corrected coefficient discussion and the comparison with the earlier Newtonian result around 01:32:35–01:36:53.

Newton's treatment did not lead us astray; it gave the same basic equation. What general relativity adds is the interpretation. From the Newtonian point of view, the extra constant distinguishes above escape velocity, below escape velocity, and the knife-edge case. From the Einstein point of view, the same sign tells us the geometry of space itself.

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[a] **Question from the audience.** How did this work before one included a cosmological constant?

Response. Before the cosmological constant, there was a tight relationship between escape-velocity behavior and spatial geometry. If $k = +1$, space is positively curved and the motion is below escape velocity, so the universe can expand and then recontract. If $k = -1$, space is negatively curved and the motion is above escape velocity, so the universe keeps expanding. If $k = 0$, one has the marginal knife-edge case.

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[a] Lecture timestamp: 01:42:46.

The same sign k can therefore be read in two ways: as a dynamical classification of the motion and as the sign of the curvature of space.

Chapter 6

Last Scattering and Baryogenesis

6.1 Looking Outward Means Looking Backward

At different distances we see the universe at different times. Nearby, the look-back time is not very important. The universe a year ago, or a hundred years ago, was not very different from the universe now. But far enough out, we see the universe younger. A younger universe is denser, and a denser universe is hotter.

At some sufficiently large distance, the universe was about as hot as the surface of the sun. It does not look that way to us because the photons have been redshifted on the way here. A redshift factor of roughly a thousand means an observed temperature lower by the same factor:

$$T_{\text{obs}} \sim \frac{T_{\text{emit}}}{1+z}, \qquad 1+z \sim 10^3.$$

[1]

That is where the universe became opaque. The universe was mainly hydrogen, with a bit of helium. Above a temperature of a few thousand degrees, hydrogen is ionized. So beyond that surface there are not neutral atoms but ions: mainly protons and electrons moving freely. The photons are also more energetic there, basically X-rays. If an atom forms accidentally, it is quickly reionized again by collisions or by a photon.

An ionized gas is an electrical conductor, and a conductor does not let radiation pass through cleanly. Charges respond to electric fields, slosh back and forth, and re-emit radiation. So beyond that distance the universe is opaque to us. The surface of last scattering is like the visible surface of the sun: not a material wall, but a place where the medium rather suddenly goes from opaque to transparent.

Most of the particles there are photons, by a factor of about 10^8. The same is true today, although today the photons are so low in energy that they are not an important part of the present energy density. Once the temperature drops below the ionization temperature of hydrogen, the universe becomes clear and we can see through it.

Direct sight ends there. The next question is how much farther back known atomic, nuclear, and particle physics can still take us.

[1] This display is a compact editorial transcription of the verbal redshift argument at 00:02:20–00:02:39.

6.2 Going Farther Back

Just inside last scattering, in addition to photons, the main particles are protons and electrons, with some helium nuclei and a few deuterons. There are practically no antiprotons and practically no positrons. But farther back, the temperature eventually becomes high enough that two photons can collide and produce an electron and a positron:

$$\gamma + \gamma \leftrightarrow e^- + e^+.$$

The basic threshold is that the photons must at least supply the rest energy of the pair:

$$E \geq 2m_e c^2.$$

Below that threshold ordinary light does not do the job. Above it, electron–positron pairs can be created, and the inverse reaction can also happen. A hot pot of material at that temperature would contain particle reactions playing the role that chemical reactions play in ordinary matter.

Just above threshold there are still more photons than electrons and positrons. Electron rest mass is expensive. But as the temperature rises far above threshold, the electron rest energy becomes less important than the kinetic energy, and the densities become comparable. Then electrons, positrons, and photons all have roughly the same density and are colliding around like mad.

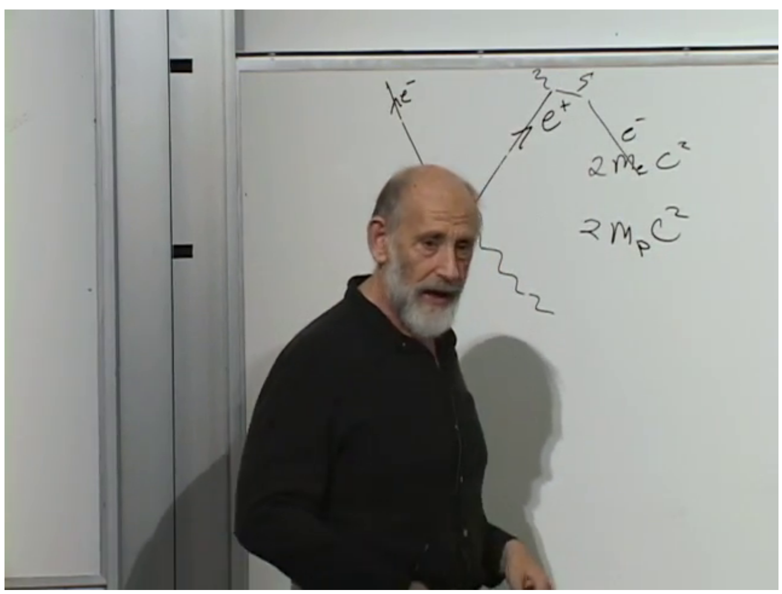

Figure 6.1: Pair-production threshold sketch comparing the electron and proton mass scales.
Lecture frame: 00:18:47.

The next question is what happens to protons. They are already present, and by then they are very dense, but there is not yet enough thermal energy to create proton–antiproton pairs in abundance. The reason is simple:

$$m_p \approx 2000\, m_e, \qquad E \geq 2m_p c^2$$

is a much higher threshold.[2]

[2]Editorial clarification: In the frame the lower mass label is less sharp than the upper $2m_e c^2$, but the spoken comparison is explicitly between the electron and proton thresholds.

This is already much hotter than the interior of a star, where antiparticles are not present in any substantial abundance.

But even that is only a provisional description. By the time the temperature is high enough to make protons and antiprotons freely, it is also high enough to smash protons into quarks. So the really hot soup is better described as a soup of electrons, positrons, quarks, antiquarks, photons, neutrinos, gluons, and other species. A simple schematic reaction from the board is

$$e^- + e^+ \to \gamma \to q + \bar{q}.$$

[a] **Question from the audience.** Is it mesons or pre-quarks?

Response. Mesons are made out of quarks. Once the temperature is high enough to shatter a proton, it also shatters the mesons.

[a]Lecture timestamp: 00:20:05.

So far the reasoning has been backward in time. Now turn it around and ask what happens as the hot symmetric soup cools.

6.3 Electrical Neutrality and a Symmetric Hot Universe

At sufficiently high temperature the universe contains huge densities of particles and antiparticles. Electrons and positrons occur in pairs, quarks and antiquarks occur in pairs, and the natural picture is nearly equal abundances of each particle and its antiparticle.

Electrical neutrality needs a little care. The universe is electrically neutral today. A useful mnemonic is to imagine, not our actual universe, but a closed and bounded one, like a three-sphere. Think first of a two-dimensional sphere as a model. Electric field lines cannot begin or end except on charge. A positive charge emits lines of force; a negative charge absorbs them. On a closed sphere those lines cannot simply escape to infinity. They refocus at the opposite side. So, as a mnemonic, a closed and bounded universe cannot carry net charge. Whether our actual universe is literally closed is not the point. The point is that large-scale positive and negative charges appear to cancel extraordinarily well.

Electrical neutrality does *not* mean that the number of protons is the same as the number of antiprotons. It means that the total positive charge equals the total negative charge. Protons, positrons, and every other positively charged species together balance electrons and every other negatively charged species together.

As far back as known physics can be trusted, and perhaps a little beyond, the universe was hot enough to sustain both particles and antiparticles in abundance. Any accelerator experiment one can name had an analogue in the early universe at some stage. A nearly symmetric start is the natural conjecture: equal numbers of electrons and positrons, equal numbers of quarks and antiquarks, and, if protons and antiprotons are used as stand-ins for baryons and antibaryons,

$$N_p = N_{\bar{p}}.$$

That conjecture is attractive because the laws of physics are almost symmetric between particles and antiparticles. A positron behaves very much like an electron, not quite but very nearly. That near symmetry is the problem.

6.4 Running the Movie Forward

Suppose the universe begins in a hot state with equal or nearly equal abundances of particles and antiparticles, and then cools.

While the temperature is high enough, protons and antiprotons can be created from energetic collisions, and they can annihilate back into photons. But eventually the temperature falls below the proton threshold. Then proton–antiproton pairs can still annihilate into photons, but after a little expansion those photons have lost enough energy that they can no longer turn back into proton–antiproton pairs. This is the one-way street. The same thing happens later for electrons and positrons, at a much lower temperature than the proton threshold but still far above stellar temperatures.

Because the expansion is slow compared with microscopic reaction rates, the universe is mostly close to thermal equilibrium while this is happening. This is what is meant here by adiabatic expansion. In that setting the more accurate statement is that the entropy changes very little. For a hot particle soup, entropy is essentially the number of particles, up to factors of order one. So the protons, antiprotons, electrons, and positrons that annihilate are roughly replaced by photons.

If the numbers of protons and antiprotons were exactly equal, they would simply annihilate away. The same would happen to electrons and positrons. One would be left with photons. But if there were even a tiny excess of protons over antiprotons, the matched pairs would annihilate and the unmatched excess would remain. A concrete example makes the point: if there were one hundred protons and fifty antiprotons, the fifty antiprotons would find fifty protons, annihilate with them, and fifty protons would remain. Charge neutrality then requires the same sort of leftover excess for electrons over positrons.

This lets one read the early asymmetry from the present photon count. Today the number of photons is about 10^9 to 10^{10} times the number of protons or electrons:

$$\frac{N_\gamma}{N_p} \sim 10^9\text{–}10^{10}.$$

So in the early hot soup there was only about one extra proton for every billion proton–antiproton pairs. That tiny imbalance is the whole reason atoms still exist.

---

[a] **Question from the audience.** Why not think of the excess statistically, as something created during cooling with a tiny probability?

Response. To get an extra proton from an initially symmetric state, one would have to create a proton without also creating an antiproton. One must also create negative charge with it, say an electron. The known standard model contains no such low-energy process. At those tested energies baryon number is conserved, so ordinary cooling does not generate the net excess.

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[a] Lecture timestamp: 00:42:25.

So electric charge is not the obstacle. A schematic example would be

$$\gamma + \gamma \to p + e^-.$$

[3] This would be electrically neutral overall. But it violates another empirical rule. In the tested standard model one cannot create a proton without also creating an antiproton. For present purposes

[3] This display is an editorial transcription of the verbal example at 00:44:16–00:46:35. Its purpose is to isolate the point that electric charge could be conserved while baryon number is violated.

the relevant conserved quantity is baryon number,

$$B = (N_p + N_n) - (N_{\bar{p}} + N_{\bar{n}}).$$

[4] At known energies that quantity is conserved, so the asymmetry cannot be manufactured during the late, well-tested part of the cooling history.

6.5 Baryogenesis

This is the subject called baryogenesis. Suppose the universe really did begin with equal numbers of baryons and antibaryons. What would the laws of physics have to allow in order to tilt that symmetric state into the tiny observed excess?

On the board this was reduced to two symbolic expressions,

$$N_p = N_{\bar{p}}, \qquad N_p - N_{\bar{p}},$$

and then to a list of conditions connecting the first to the second.

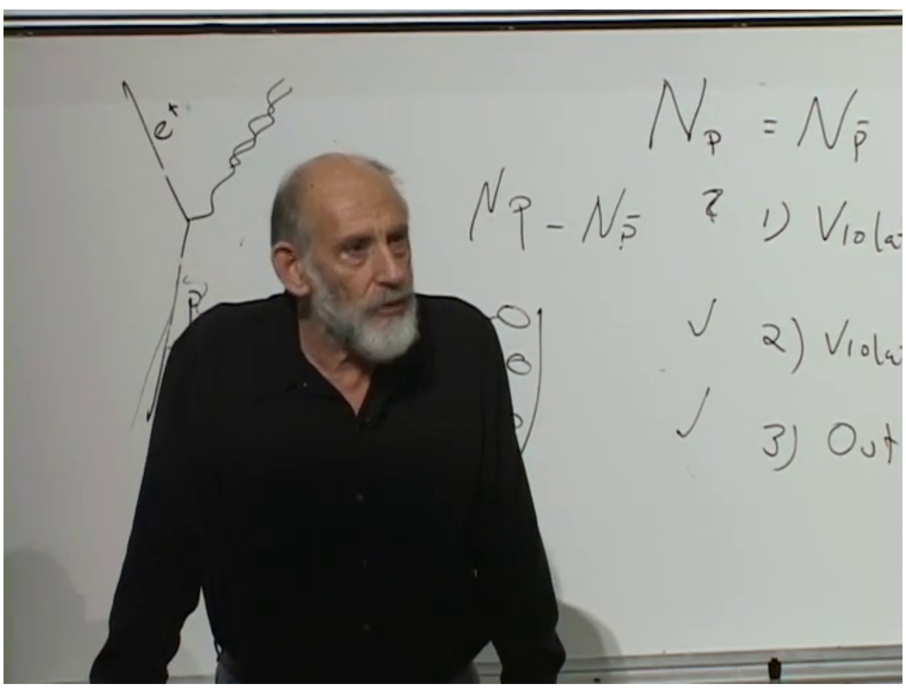

Figure 6.2: Conditions for generating a baryon asymmetry from an initially symmetric state.

Lecture frame: 01:22:26.

[5]

The first condition is violation of baryon number. If baryon number is an exact conservation law, a universe that starts with zero net baryon number stays that way.

[4] This compact formula formalizes the verbal definition given at 00:45:18–00:45:40.

[5] Editorial clarification: In this frame the words in the numbered list are only partly legible. The three items below follow the accompanying spoken discussion.

The second condition is a bias between particles and antiparticles. Baryon-number violation alone is not enough. If nature allows a proton-producing process and its reflected antiproton-producing process with equal probability, the average imbalance is still zero. One needs the rates to differ.

Ordinary statistical fluctuations are also far too small. If the total number of particles is about 10^{90}, including photons, then a fluctuation of order $\sqrt{N}$ gives only about 10^{45}, nowhere near enough to explain the roughly 10^{80} protons in the universe.

The third condition is that the universe must at some point have been sufficiently out of equilibrium. Here the argument uses the CPT theorem. If a hot gas has sat in equilibrium forever, a movie run backward looks the same as a movie run forward. In that setting the symmetry between particles and antiparticles is effectively restored by the exact symmetry that combines charge conjugation, parity, and time reversal. To get a real bias frozen in, the actual history has to look different going forward and backward on microscopic time scales. Early enough, the expansion of the universe was fast enough to do that.

So the three conditions are:

1. violation of baryon number;
2. violation of the symmetry between particles and antiparticles, more precisely CP violation;
3. a stage out of thermal equilibrium.

---

[a] **Question from the audience.** Matching these to CPT, is condition 2 really CP and condition 3 really the time-direction issue?

Response. Yes. Condition 2 should really be stated as CP violation. Condition 3 is the fact that the actual history was out of equilibrium, not a separate conservation law and not the same thing as violating baryon number.

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[a] Lecture timestamp: 01:09:08.

With these three conditions in place, even a universe that starts with equal numbers of protons and antiprotons can be tilted one way or the other. What remains uncertain is whether nature really violates baryon number, and if so how.

6.6 What Is Known, and What Is Not

The open part of the story is baryon-number violation. We have never directly seen a reaction in nature that changes

$$(N_p + N_n) - (N_{\bar{p}} + N_{\bar{n}}).$$

That may only mean that the relevant reactions require much higher temperatures than anything directly tested in the laboratory, or that they are fantastically improbable at low energy. But at present this is still the question mark.

That is why proton decay matters so much in the discussion. If baryon number is not exact, then the proton should in principle be unstable. But it is spectacularly stable. The lower bound quoted here is of order

$$\tau_p \gtrsim 10^{34} \text{ years}.$$

This comes from huge experiments using very large volumes of water instrumented with light detectors and buried deep underground. It does not disprove baryon-number violation. It means only that any such violation must be extraordinarily suppressed at low energy.

A separate known-physics remark also enters the discussion. Around 100 to 250 GeV, the weak interactions become much more like the electromagnetic interactions. That may matter in detailed models, but it is not the dramatic step in the present argument.

Dark matter does not change the basic logic much. In most theories it interacts so weakly with ordinary matter that, for this question, it is essentially a spectator.

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[a] **Question from the audience.** How does gravity affect all this?

Response. Mainly in two ways. First, without cosmic expansion there would be no out-of-equilibrium stage. Second, black-hole evaporation gives a strong reason to suspect that baryon number cannot be an exact conservation law forever. Apart from that, ordinary gravitational interactions between particles are too weak to matter much here.

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[a]Lecture timestamp: 01:24:49.

There is then a follow-up question about electric charge. Would the same black-hole argument also threaten charge conservation? The answer is no. A charged black hole carries electric flux lines extending outward, and those lines keep track of the charge. Such a black hole does not evaporate all the way away; it approaches an extremal remnant instead. So the black-hole argument threatens baryon number, not electric charge.

Another question asks whether the very dense early quark soup should simply be regarded as a black hole. The answer is no. The point is not a static dense lump. The universe was expanding. The black-hole story is a thought experiment about the fate of baryon number, not a literal description of the early universe as a trapped region.

---

[a] **Question from the audience.** Would one know how to amend the standard model to include such reactions?

Response. The trouble is the opposite. It is easy to add too much baryon violation and make the proton decay far too quickly. The real problem is keeping the proton as stable as it is while still allowing the early-universe processes one wants.

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[a]Lecture timestamp: 01:31:14.

6.7 Entropy and the Late Neutron Questions

Return now to the bookkeeping. In an adiabatically expanding universe, entropy is approximately conserved. For a hot soup of elementary particles, entropy is essentially the particle number up to factors of order one. That is why the discussion does not need a huge late production of photons out of nowhere. The annihilations mostly reshuffle the content of the soup into radiation.

The older version of the problem had the story upside down. Instead of asking why so few protons survived relative to photons, the question was asked the other way around: why are there so many photons compared with protons? But the sharper question is not why the universe manufactured so much entropy late in the game. It is why the asymmetry between particles and antiparticles was so tiny to begin with: only about one extra proton per billion proton–antiproton pairs.

A free neutron does decay, but that process does *not* violate baryon number:

$$n \to p + e^- + \bar{\nu}.$$

The neutron carries the same baryon number as the proton.

For a free neutron the available decay energy is very small, because the neutron mass is only slightly above the mass of the final state. That is part of why the decay is so slow by elementary-particle standards. When a neutron is bound in a nucleus, binding energy can remove that small amount of available energy altogether, and then the bound neutron can be stable.

[a] **Question from the audience.** Are neutrons simply disappearing from the universe, or are they also created?

Response. The first answer is that free neutrons are lost through decay. But that is not the whole story. There are real sources of neutron production, including nuclear reactions in stars and the formation of neutron stars. So the total neutron count is not simply a one-way decrease.

[a] Lecture timestamp: 01:43:03.

Chapter 7

Vacuum Energy, Inflation, and Primordial Fluctuations

7.1 Vacuum Energy and de Sitter Space

In flat space the Friedmann equation is

$$\left(\frac{\dot{A}}{A}\right)^2 = \frac{8\pi G}{3}\rho. \tag{7.1}$$

For matter, ρ decreases like A^{-3}. For radiation, ρ decreases like A^{-4}. For vacuum energy it does not dilute:

$$\rho = \rho_0 = \text{constant}. \tag{7.2}$$

If the universe contains nothing but this vacuum energy, the right-hand side is a constant. Writing that constant as H^2,

$$H^2 = \frac{8\pi G}{3}\rho_0, \qquad H = \sqrt{\frac{8\pi G}{3}\rho_0}. \tag{7.3}$$

Then the Hubble constant is truly constant:

$$\frac{\dot{A}}{A} = H. \tag{7.4}$$

So

$$\dot{A} = HA, \tag{7.5}$$

and the solution is exponential:

$$A(t) = Ce^{Ht}. \tag{7.6}$$

The distance between distant unbound galaxies grows exponentially. This is de Sitter space.

In these coordinates the geometry is

$$d\tau^2 = dt^2 - e^{2Ht}\left(dx^2 + dy^2 + dz^2\right). \tag{7.7}$$

The vacuum energy here is positive. In flat space it has to be, because the left-hand side of the Friedmann equation is nonnegative. A negative vacuum energy can appear only when curvature terms are included, and that is not the case being followed here.

Einstein introduced the cosmological constant for a different purpose: to balance ordinary matter and hold the universe static.

7.2 The Present Accelerating Universe

De Sitter space is also the late-time tendency of the present universe. Earlier the scale factor followed the matter-dominated behavior

$$A(t) \propto t^{2/3}. \tag{7.8}$$

A few billion years ago the curve bent upward and began to increase more rapidly. It is not yet a perfect exponential, because other forms of energy are still present, but the trend is toward one.

Acceleration here means

$$\ddot{A} > 0. \tag{7.9}$$

This is an observed feature of the late-time expansion.

Now consider the horizon. Hubble's law says

$$v = HD. \tag{7.10}$$

At the distance where the recession speed becomes the speed of light,

$$D = \frac{c}{H}. \tag{7.11}$$

Equivalently, one can rewrite this as $HD = c$.[1]

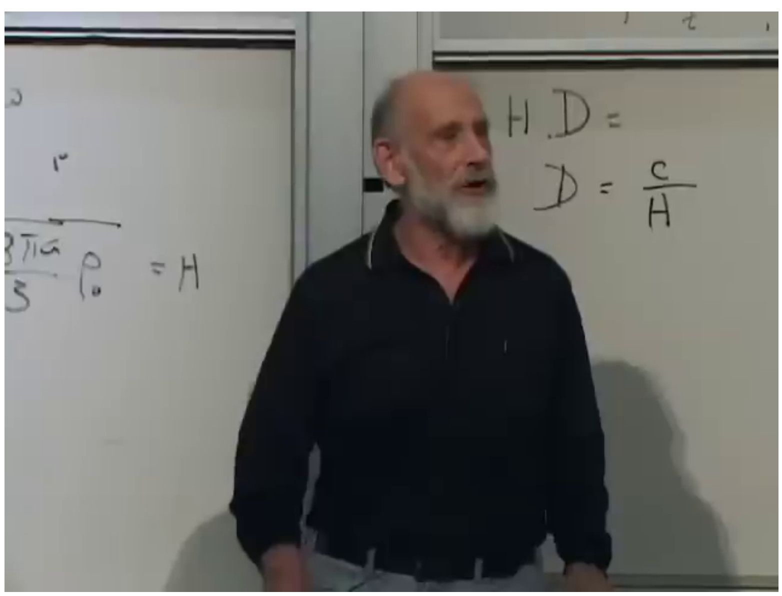

Figure 7.1: Cosmic horizon relation $D = c/H$ in de Sitter expansion.

Lecture frame: 00:30:37.

[1] Editorial clarification: This is an editorial algebraic restatement of the spoken relation $D = c/H$. The upper line on the right board in Figure 7.1 is only partly legible.

This distance is of order ten to fifteen billion light years, numerically close to the age of the universe once the latter is converted to a distance with c.

---

[a] **Question from the audience.** Isn't the near equality between the horizon scale and the age of the universe just what happens if the tangent line happens to meet the origin?

Response. That would be exactly true for $A(t) \propto t$. It is not true in general. In de Sitter space H stays fixed while the age keeps increasing, so the present near equality is not a mathematical identity.

---

[a]Lecture timestamp: 00:14:10.

7.3 What Vacuum Energy Means

Vacuum energy is described here in the language of ordinary quantum mechanics. For a harmonic oscillator the energy is schematically

$$E \sim p^2 + x^2. \tag{7.12}$$

Classically the ground state is the equilibrium point,

$$x = 0, \qquad p = 0, \qquad E = 0. \tag{7.13}$$

Quantum mechanics does not allow both x and p to be exactly fixed at zero. The uncertainty principle forces fluctuations, so the ground-state energy is positive. For the oscillator one gets

$$E_0 = \frac{1}{2}\hbar\omega. \tag{7.14}$$

The same point is then carried over to fields. The universe is full of harmonic oscillators: the oscillations of the electromagnetic field, and likewise of every other field. Each field has zero-point energy. For the electromagnetic field the energy density is schematically

$$u_{\mathrm{EM}} \sim E^2 + B^2. \tag{7.15}$$

Even with no real photons present there remain irreducible zero-point oscillations. That is one explanation of vacuum energy. The only property needed here is that this energy is present and does not dilute as the universe expands.

---

[a] **Question from the audience.** What is this vacuum energy, and why would anything remain if matter and radiation were removed?

Response. Because the vacuum is the ground state of the quantum fields, and a quantum ground state need not have zero energy. Harmonic oscillators have positive ground-state energy, and fields behave as collections of oscillators. So even with no real particles present there remains zero-point energy.

---

[a]Lecture timestamp: 00:16:15.

This is not the three-degree background radiation. The microwave background is ordinary radiation. Its energy density decreases because the photons dilute and each photon redshifts. Vacuum energy is the residual zero-point energy that remains after the real photons have been removed.

7.4 What H^{-1} Means

Now return to the relation between the inverse Hubble constant and the age of the universe. If

$$A(t) \propto t, \tag{7.16}$$

then

$$H = \frac{\dot{A}}{A} = \frac{1}{t}. \tag{7.17}$$

More generally, if

$$A(t) \propto t^p, \tag{7.18}$$

then

$$H = \frac{p}{t}. \tag{7.19}$$

For the matter-dominated case $p = 2/3$,

$$H = \frac{2}{3t}. \tag{7.20}$$

So in power-law cosmologies H^{-1} is of the same order as the age, but not exactly the age.

De Sitter space is different. There

$$A(t) = Ce^{\gamma t}, \tag{7.21}$$

and therefore

$$H = \frac{\dot{A}}{A} = \gamma, \tag{7.22}$$

a true constant. Here H^{-1} is not the age of the universe. It is the time constant for the exponential growth: the time in which the scale factor is multiplied by e. The spoken discussion repeatedly says "double" and then corrects it; the precise statement is multiplication by e. Today that time constant is of order ten billion years.

---

[a] **Question from the audience.** Could you go over the point about the age of the universe and the inverse Hubble constant?

Response. For power laws one gets $H = p/t$, so H^{-1} tracks the age up to a factor of order one. But in de Sitter space H is constant while the age keeps growing, so H^{-1} is only the exponential time constant, not the age.

---

[a]Lecture timestamp: 00:23:46.

7.5 Energy and the Cosmic Horizon

If the vacuum-energy density stays constant while the universe expands, it is natural to ask whether the total vacuum energy is increasing. The answer given here is limited. In general relativity the energy bookkeeping includes a gravitational component, so the notion of total energy is subtler than it is in elementary mechanics. The practical point is the one tied to the horizon.

In exact de Sitter space the cosmic horizon has fixed size,

$$D = \frac{c}{H} = \frac{c}{\sqrt{8\pi G\rho_0/3}}. \tag{7.23}$$

The volume inside that horizon is therefore fixed. The vacuum energy within it is

$$E_{\rm hor} = \frac{4\pi}{3} D^3 \rho_0. \tag{7.24}$$

[2] Since both D and ρ_0 are constant, this quantity does not change with time. For the portion of the universe that can ever be seen by a given observer, the accessible vacuum energy stays fixed in exact de Sitter space.

Things can also pass through the cosmic horizon in much the same sense that they pass through a black-hole horizon. Once outside, they are out of causal contact with us.

---

[a] **Question from the audience.** If the vacuum-energy density stays constant while the universe expands, where does the extra energy come from?

Response. The lecture answers in general-relativistic language, not Newtonian box language. In exact de Sitter space the horizon size stays fixed, so the vacuum energy inside the region that can ever be seen stays fixed as well.

---

[a]Lecture timestamp: 00:29:27.

7.6 Why Inflation Was Introduced

The same de Sitter mechanism is then pushed into the remote past, but with a much shorter time constant, of order

$$H^{-1} \sim 10^{-32}\,\text{s}. \tag{7.25}$$

That is the idea of inflation: before anything directly observable, the universe underwent a period of exponential expansion.

The basic motivation is the flatness and smoothness of the universe. The spatial slices appear very flat. On large scales the universe is also very homogeneous. More strikingly, the surface of last scattering was already smooth, so the smoothing could not have happened only during the later visible history.

A deflated balloon is wrinkled. Blow it up enough and any visible patch becomes smooth. The raisin or prune analogy makes the same point: a wrinkled object, stretched by an enormous factor, looks smooth on any visible patch. Inflation is supposed to have done exactly that.

This also answers the horizon problem in the form used here. Widely separated regions of the last-scattering surface are both visible to us, but light did not have time to travel from one to the other during the later visible history. Their nearly uniform density and temperature therefore suggest a prehistory in which one smaller region was stretched enormously.

Inflation has a mixed virtue. It wipes out sensitivity to the initial conditions, and for the same reason it wipes out much of the information about how the universe started.

The flatness question reappears here in sharper form. Observationally the curvature parameter k looks like zero, at the level of about one percent or better. Geometry looks Euclidean over the visible region. Inflation explains this by making the radius of curvature much larger than the region we can see.

[2]Editorial clarification: This condenses the spoken calculation at 00:31:06–00:32:23: take the horizon size, form the spherical volume, and multiply by ρ_0.

[a] **Question from the audience.** Are we assuming equilibrium before inflation?

Response. No. The point of inflation is almost the opposite. Whatever structure was there beforehand gets stretched so much that the visible region loses memory of the earlier details.

[a]Lecture timestamp: 00:40:05.

7.7 A Scalar Field and Its Potential

To explain how the effective vacuum energy could have been much larger in the early universe than it is now, introduce a scalar field. Unlike the electric and magnetic fields, which are vectors, a scalar field has no direction. It simply has a value at each point:

$$\phi = \phi(x, y, z, t). \tag{7.26}$$

This field is hypothetical. It is not identified with any known field in nature, and its quanta are not identified with any known particles.

The part of its energy that matters here is a potential energy,

$$V(\phi). \tag{7.27}$$

By "potential energy" one means energy associated with the value of the field itself, not with the time derivative of the field. Different values of ϕ can have different energies.

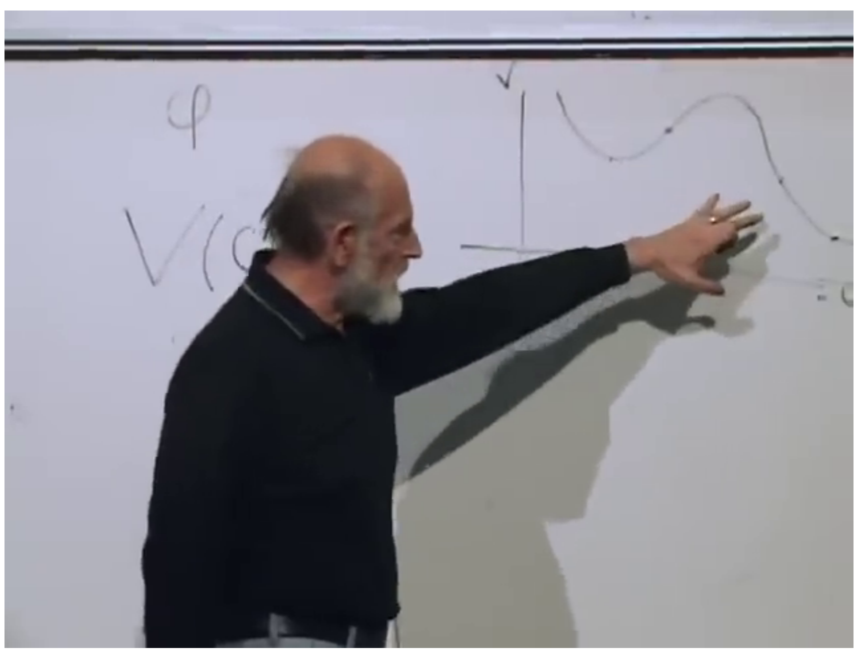

Figure 7.2: Qualitative scalar potential $V(\phi)$ showing that different field values carry different potential energies.

Lecture frame: 00:47:08.

[a] **Question from the audience.** What is the horizontal axis in the $V(\phi)$ drawing?

Response. The horizontal axis is the field value ϕ, and the vertical axis is the potential energy V.

[a]Lecture timestamp: 00:46:34.

If the field is constant in space and time, or nearly constant over the region of interest, then $V(\phi)$ behaves exactly like vacuum energy. It does not dilute. So the earlier de Sitter formulas are reused with ρ_0 replaced by $V(\phi)$. In particular,

$$H = \sqrt{\frac{8\pi G}{3} V(\phi)}. \tag{7.28}$$

If ϕ varies from place to place, different regions inflate with different time constants. But if the stretching is large enough, any visible patch becomes nearly homogeneous.

[a] **Question from the audience.** Is ϕ acting like a time parameter here?

Response. In this discussion it does, in the limited sense that while the field rolls down the potential it is a monotonic function of time.

[a]Lecture timestamp: 01:05:11.

7.8 A Plateau, a Cliff, and Reheating

The potential needed for this story becomes very flat, though not perfectly flat, then reaches the edge of a cliff, drops sharply, does not quite go to zero, and rises again.

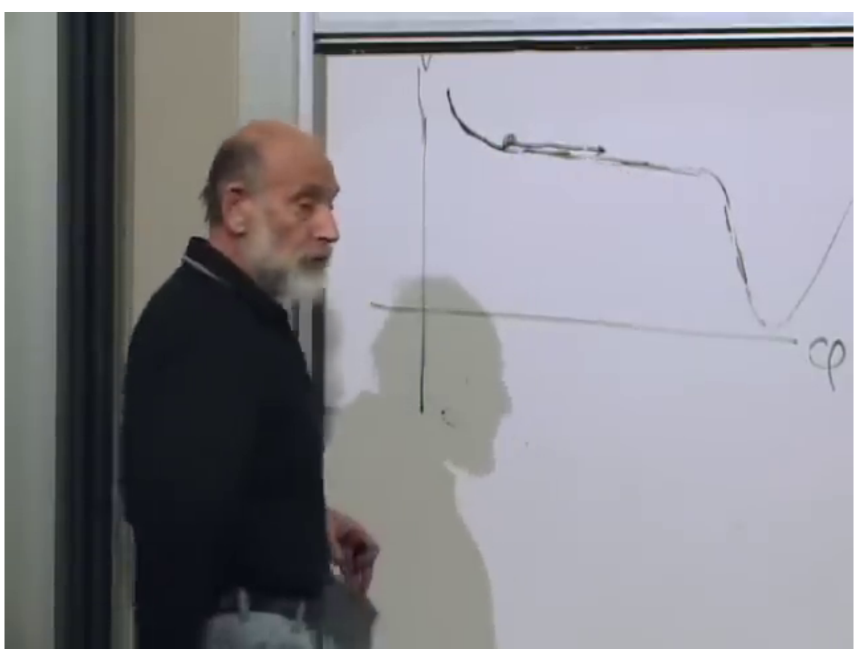

Figure 7.3: Inflationary potential with a high plateau, a steep drop, and a lower minimum.

Lecture frame: 01:02:30.

This shape is highly contrived, but it is introduced because the cosmological story seems to require something like it.

The field begins high on the plateau. Because the slope is small and the effective viscosity associated with the expansion is large, ϕ changes only very slowly. A rock falling slowly through honey is closer to the intended picture than an object sliding on a table. The faster the universe expands, the larger this viscosity becomes.

---

[a] **Question from the audience.** What is causing ϕ to move?

Response. The slope of the potential is pushing it. The slope is small, and the cosmological viscosity is large, so the drift is slow rather than rapid.

---

[a]Lecture timestamp: 00:59:07.

While ϕ inches along the plateau, $V(\phi)$ changes very little. For that reason the universe inflates almost as if the vacuum energy were constant. The time constant during this stage is of order

$$H^{-1} \sim 10^{-32}\,\mathrm{s}. \tag{7.29}$$

A rough spoken estimate puts the plateau height near 10^{-14} in Planck units.[3]

How long did inflation last? That is not known, because the width and shallowness of the plateau are not known. What is known is that the universe was multiplied by e at least about fifty times, and sixty is the number usually quoted.[4] These are e-foldings: each e-fold is one multiplication by e.

When the field reaches the edge, the potential energy changes rapidly. The field falls, its potential energy is converted first into kinetic energy and then into heat and particles, and the universe becomes hot. The name "reheating" is slightly misleading in this description, because any earlier thermal energy would already have been diluted away during inflation.

The lower minimum is not at zero. The small value left there is what is associated with the present slow exponential expansion.

---

[a] **Question from the audience.** Could the plateau be exactly flat, with perturbations somehow knocking the field off the edge?

Response. No. It cannot be completely flat. It has to have some tilt. If it were too flat, the reheated universe would come out too inhomogeneous.

---

[a]Lecture timestamp: 01:19:48.

"Big Bang" is used here in two nearby senses. It can mean the reheating event itself. It can also mean the earliest directly visible stage, namely the surface of last scattering. The stable point is that everything actually observed lies far below the inflationary plateau.

---

[a] **Question from the audience.** Is the reheating itself the Big Bang?

Response. If you like, yes. But the universe was still extremely hot and opaque at that point. The surface of last scattering comes much later, after the universe has cooled enough to become transparent.

---

[a]Lecture timestamp: 01:45:42.

[3]Editorial clarification: This is the spoken "best bet" estimate at 01:14:38–01:14:47, not a derived value in the lecture.

[4]Editorial clarification: Editorial restatement of the spoken e-folding discussion at 01:14:05–01:19:27.

7.9 Quantum Fluctuations and the Seeds of Galaxies

Inflation makes the visible universe very homogeneous, but not perfectly homogeneous. With no quantum fluctuations the field would become completely smooth. With quantum mechanics it still varies slightly from place to place.

The central picture plots ordinary space vertically and the field value ϕ horizontally. If the field were exactly homogeneous, every point in space would have the same ϕ at a given time, and every point would drift together toward the edge. Then every region would fall over the cliff at the same moment. Quantum fluctuations spoil that exact lockstep.

Two pictures make the point. One is a small bump in an otherwise homogeneous field. The other is a row of soldiers walking arm in arm toward a cliff. If they are perfectly aligned, they all fall together. If some are slightly ahead and some slightly behind, they go over at different times.

Now comes the important effect. A region that falls over earlier reheats earlier. By the time a later region falls over, the earlier one has already had time to expand and cool somewhat. That difference produces variations in the energy density after reheating.

---

[a] **Question from the audience.** Are the later reheated regions hotter or cooler than the earlier ones?

Response. The earlier ones have already had time to expand and cool. The later ones are therefore hotter when they fall over, and that difference contributes to the density inhomogeneity.

---

[a]Lecture timestamp: 01:34:00.

The observed effect is small:

$$\frac{\delta\rho}{\rho} \sim 10^{-5}, \tag{7.30}$$

about one part in one hundred thousand. These are primordial density fluctuations, not dark energy. This number is not derived here from the qualitative sketch alone; it is used to constrain how high and how flat the plateau can be.

They are also necessary. Without them there would be no galaxies. After reheating, gravity amplifies the initial pattern. Over-dense regions attract matter from under-dense regions. The over-dense regions become more over-dense, the under-dense ones more under-dense, and the result is the later structure of galaxies and stars.

This is not only a qualitative story. One can calculate the quantum fluctuations of a field in an expanding universe, follow them through reheating, and compare the result with the fluctuation pattern reconstructed from the galaxy distribution. The fit is very good.

The same discussion also leads to a point about entropy. Gravitational clumping looks counterintuitive, but it does not violate the second law. Gravity is peculiar in statistical mechanics: it tends to make things clump, and that clumping raises the entropy rather than lowering it.

---

[a] **Question from the audience.** How is the microwave background reflected in all of this, given that ϕ is not an electromagnetic field?

Response. The primordial density pattern becomes imprinted on the surface of last scattering. Reheating converts the field energy into ordinary particles and photons, and the microwave background later carries the imprint of those primordial fluctuations.

---

[a]Lecture timestamp: 01:40:24.

7.10 Acoustic Waves and the Next Step

At the end the discussion turns briefly to acoustic waves. These come before the later gravitational clumping. The primordial fluctuations are processed into acoustic oscillations, and those oscillations leave lumps of known physical size on the sky.

That matters because a known physical size at a known distance gives a measuring rod. Once one measures the angle it subtends on the surface of last scattering, geometry can be used to tell whether space is flat, positively curved, or negatively curved. The details are postponed.

---

[a] **Question from the audience.** What about the acoustic waves?

Response. They provide a measuring rod on the sky. Their known physical size, together with their observed angular size on the surface of last scattering, is what lets one measure the geometry of space. The details are deferred to the next lecture.

---

[a]Lecture timestamp: 01:47:19.

Chapter 8

Inflation, Structure, and the Loss of Initial Conditions

8.1 Omega and the Evidence for Flatness

A student's question about big Ω sends the discussion back to the Friedmann-Robertson-Walker equation. For the present era, well after radiation domination, the expansion law is written with a vacuum-energy term and a matter term, while radiation is dropped because it is now very small:

$$H^2 = \left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\left(\rho_0 + \rho_m\right) - \frac{k}{a^2}. \tag{8.1}$$

The vacuum term does not dilute. The matter term does:

$$\rho_m = \frac{M}{a^3}. \tag{8.2}$$

Ω is the fraction of the expansion equation carried by the different components.

Divide by H^2. Then the normalized terms add to one. In a regularized notation,

$$\begin{aligned} \Omega_0 &= \frac{8\pi G\rho_0}{3H^2}, \\ \Omega_m &= \frac{8\pi G\rho_m}{3H^2}, \\ \Omega_k &= -\frac{k}{a^2H^2}, \end{aligned} \tag{8.3}$$

and therefore

$$\Omega_0 + \Omega_m + \Omega_k = 1. \tag{8.4}$$

[1] If space is flat, $k = 0$, so

$$\Omega_0 + \Omega_m = 1. \tag{8.5}$$

If the matter-plus-vacuum sum is greater than one, the curvature is positive. If it is less than one, the curvature is negative.

[1] Editorial clarification: The audio and blackboard are rough in this stretch, and the speaker shifts among several names for the vacuum contribution. The notation here regularizes the argument after division by H^2.

The observational logic is then straightforward. Measure ordinary matter and dark matter. Measure the Hubble expansion. Measure the curvature geometrically on very large scales. Notice also that the universe is accelerating. The sign of the vacuum term is not H by itself but the tendency of $\dot{a}/a$ toward a constant, which means that the scale factor approaches the exponential branch,

$$a(t) \propto e^{Ht}. \tag{8.6}$$

The several measurements cross-check one another and say that the universe is very nearly flat, to within a few percent.

---

[a] **Question from the audience.** Does matter include dark matter?

Response. Yes. For the present discussion the matter term includes dark matter. One could also add radiation, but today that fraction is so small that it can be ignored.

---

[a]Lecture timestamp: 00:10:50.

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[a] **Question from the audience.** Is there some measure of dark energy other than the Hubble constant?

Response. The point is not the value of H by itself, but the way the expansion tends toward an exponential law. Together with the matter measurement and the geometric measurement of flatness, that lets one infer the vacuum term.

---

[a]Lecture timestamp: 00:11:11.

8.2 From Flatness to Structure

Why is the universe so flat? The proposal is inflation. Flat means very big, very smooth, and very homogeneous on scales of billions, really tens of billions, of light years. Exponential expansion can stretch a region until it looks that way.

But one can overdo a good thing. A perfectly smooth universe stays perfectly smooth. Gravity does not manufacture lumpiness out of exact symmetry. If structure exists now, then structure had to begin from a small inhomogeneity earlier. Tracing the present universe backward, the inhomogeneity at the time of transparency comes out at roughly one part in 10^5:

$$\frac{\delta\rho}{\rho} \sim 10^{-5}. \tag{8.7}$$

The universe was extremely smooth, but not perfectly so.

From there gravity does what ordinary forces in a room full of air do not do. A small overdensity pulls matter toward itself. The nearby region is depleted, so the underdensity becomes more underdense and the overdensity becomes more overdense. Gravity reinforces the contrast instead of erasing it.

The natural question is why this did not happen earlier. The answer is pressure. Earlier, radiation pressure was large. Using

$$p = w\rho, \tag{8.8}$$

the familiar cases are

$$w_{\text{rad}} = \frac{1}{3}, \qquad w_m = 0, \qquad w_{\text{vac}} = -1. \tag{8.9}$$

Positive pressure homogenizes. It pushes disturbances out and smooths the medium. Matter, treated on large scales as pressureless, offers no such support. With pressure present, gravity is held off. When pressure becomes negligible, gravity wins.

Dark matter is crucial in the order of events. The first collapsing lumps are dark-matter halos. Ordinary baryonic matter then falls into those halos and, because baryons can dissipate, collects nearer the center. The dark matter does not dissipate in the same way, so it remains spread through the halo. The beginning of that contraction is slightly earlier than transparency, by roughly a factor of ten in time, which is why the microwave background does not show the onset directly.

A brief aside fixes the later chemical history. Almost all elements heavier than helium are products of stellar evolution rather than early-universe cosmology. The early universe makes essentially hydrogen, helium, and small traces of a few light nuclei.

What is observed on the sky is not mass density itself but temperature anisotropy in the microwave background. Density fluctuations contribute to it. Velocity fluctuations contribute too. Overdense regions tend to look slightly colder because photons lose extra energy climbing out of deeper gravitational wells, while Doppler shifts from the motion of the gas are mixed in on top of that. The full numerical analysis is complicated. The robust statement is simpler: by the time the universe became transparent, the energy density varied by about one part in 10^5.

[a] **Question from the audience.** Why didn't gravity make the lumps grow earlier?

Response. Because earlier the universe was radiation dominated, and radiation carried enough positive pressure to homogenize the medium. Only when radiation became unimportant compared with matter did gravity begin to dominate the growth of the lumps.

[a] Lecture timestamp: 00:20:29.

[a] **Question from the audience.** Don't you need a dissipation mechanism for collapse?

Response. Ordinary baryonic matter does need dissipation if it is to settle toward the center. But dark matter can begin by making halos without that sort of collisional dissipation. There is also a cosmological kind of damping tied to expansion itself, and that will matter later.

[a] Lecture timestamp: 00:24:35.

[a] **Question from the audience.** Were there also velocity fluctuations, so that later galaxies could acquire angular momentum?

Response. Yes. The primordial pattern involved density fluctuations and velocity fluctuations. What is seen on the sky is a temperature pattern produced by both the density variations and the motion of the gas.

[a] Lecture timestamp: 00:30:34.

8.3 The Inflaton and the Plateau

The mechanism now introduced is a scalar field ϕ. Its energy density has three pieces:

$$\rho_\phi = \frac{1}{2}\dot{\phi}^2 + \frac{1}{2}(\nabla\phi)^2 + V(\phi). \tag{8.10}$$

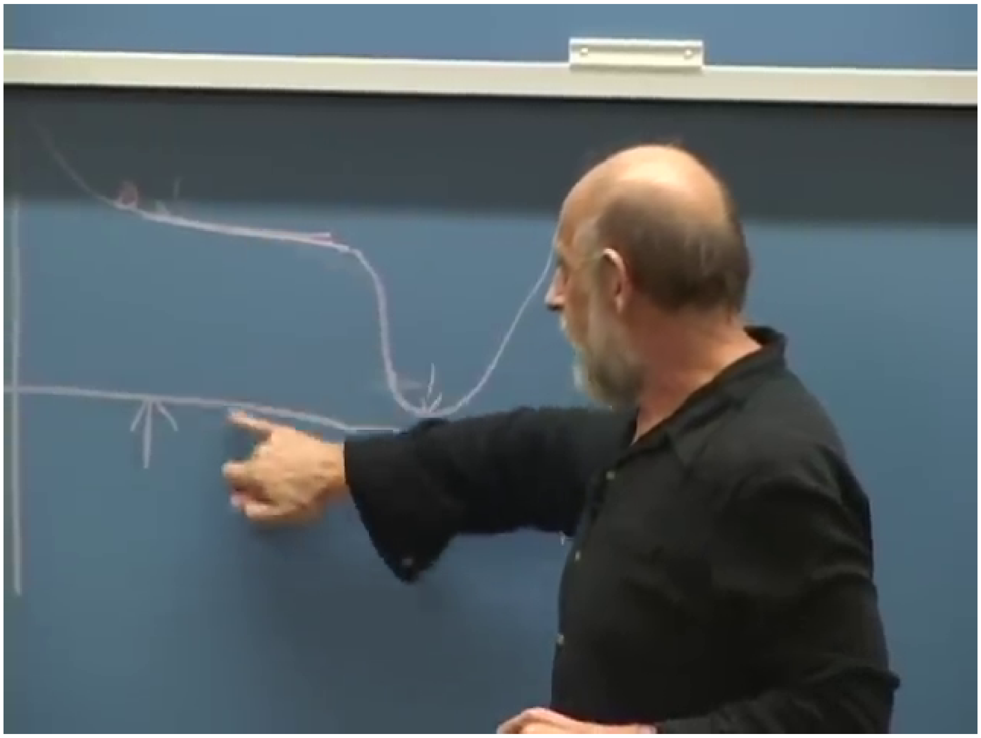

Figure 8.1: Qualitative inflaton potential with a broad plateau, a steep drop, and a low minimum.
Lecture frame: 00:43:23.

Time variation contributes. Spatial variation contributes. A scalar field can also carry a true potential energy $V(\phi)$. That part is not exotic; the Higgs field is the familiar example. What matters here is the shape of the potential.

The potential must be rather flat over some range, with a slight downward slope, and then at a critical value the bottom drops out. It does not fall to exactly zero. It falls to something very small. In Planck units the present vacuum energy is of order

$$10^{-123}. \tag{8.11}$$

Up on the plateau it was much larger than this, but still far below the Planck scale.

This plot does not describe a place in space. It describes a value of the field. To say that the universe started high on the plateau is only to say that the field value, over a very large region, lay up there. It is not important if ϕ varied somewhat from place to place at first, because inflation stretches those variations enormously. After enough expansion, any local patch looks almost uniform. [2]

As long as the field sits on the broad, slowly sloping part of the potential, the potential energy behaves like vacuum energy. If ϕ were perfectly frozen, it would behave exactly like vacuum energy. Leaving the omitted numerical factors aside, the relation stated in the lecture is simply

$$H \propto \sqrt{V(\phi)}, \tag{8.12}$$

and that gives the nearly exponential law

$$a(t) \propto e^{Ht}. \tag{8.13}$$

[2] Editorial clarification: The axis labels are not legible in the blackboard frame, but the spoken discussion makes clear that the vertical axis is the potential energy $V(\phi)$ and the horizontal axis is the field value ϕ.

Almost everything that can now be observed happens after the field goes over the edge. The drop is where the large vacuum-like energy on the plateau is converted into radiation, ordinary matter, dark matter, and the tiny leftover vacuum energy seen today.

---

[a] **Question from the audience.** What is ϕ a field on, and are Einstein's equations still being used?

Response. Yes. ϕ is a field on spacetime, and the same Friedmann equation is still in use. In this regime the energy density on the right-hand side is supplied mainly by $V(\phi)$, provided the field changes slowly enough.

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[a] Lecture timestamp: 00:44:07.

---

[a] **Question from the audience.** How can vacuum-like inflaton energy turn into matter and radiation?

Response. When the field rolls to the bottom and oscillates, those oscillations are the quanta of the field. If the quanta are unstable, they can decay into photons, electrons, positrons, and other familiar particles.

---

[a] Lecture timestamp: 00:47:04.

8.4 Slow Roll and Hubble Friction

Once the field goes over the edge, why should it not simply slosh forever? Why should it not swing back and forth and perhaps even climb again? The answer given here is a cosmological damping mechanism. It is not genuine microscopic friction, but it behaves that way in the equation of motion.

Ignore spatial gradients and treat the field as homogeneous. Then the kinetic term and the potential term look like a single degree of freedom with $T-V$ dynamics, but because they are energy densities they must be multiplied by the physical volume a^3. One is led to a homogeneous effective Lagrangian of the form

$$L_{\mathrm{hom}} = a^3(t)\left[\frac{1}{2}\dot{\phi}^2 - V(\phi)\right]. \tag{8.14}$$

[3]

The equation that is secure on the board is

$$\frac{d}{dt}\left(a^3\dot{\phi}\right) = -a^3\frac{\partial V}{\partial \phi}. \tag{8.15}$$

Expanding the time derivative gives

$$a^3\ddot{\phi} + 3a^2\dot{a}\,\dot{\phi} = -a^3\frac{\partial V}{\partial \phi}, \tag{8.16}$$

and division by a^3 yields

$$\ddot{\phi} + 3H\dot{\phi} = -\frac{\partial V}{\partial \phi}. \tag{8.17}$$

The minus sign is corrected explicitly in the spoken derivation: the force is minus the derivative of the potential energy.

[3] Editorial clarification: The top line on the board is partially cropped. This homogeneous Lagrangian is reconstructed only to the extent needed to reproduce the secure equation of motion written below it.

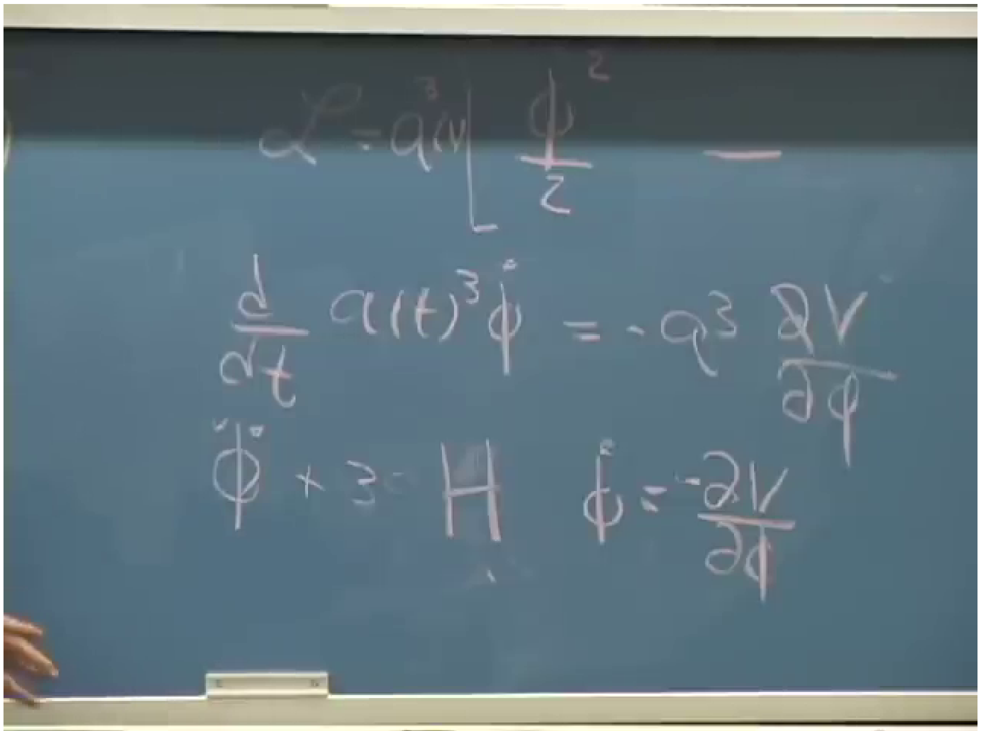

Figure 8.2: Homogeneous inflaton equation of motion showing the Hubble-friction term $3H\dot{\phi}$.
Lecture frame: 00:57:10.

The term $3H\dot{\phi}$ is what is called Hubble friction. It comes entirely from differentiating the factor a^3. Dynamically, however, it looks like the drag term in the motion of a rock falling through water. There is a force term, an acceleration term, and a term linear in the velocity. The field quickly reaches a terminal velocity and then moves slowly along the plateau. That is the slow-roll regime. While the slope is shallow and the field moves slowly, the potential energy changes only slowly, so the expansion stays close to exponential.

Near the minimum the field oscillates. Those oscillations are the inflaton quanta. If the quanta are unstable, they decay, and that is how the energy on the plateau is turned into the hot matter and radiation content of the later universe.

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[a] **Question from the audience.** Why doesn't the field just oscillate forever, and what provides the damping?

Response. Expansion itself provides a friction-like term. The factor a^3 in the homogeneous Lagrangian produces the $3H\dot{\phi}$ term, and that behaves like viscous drag.

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[a]Lecture timestamp: 00:50:48.

8.5 Quantum Fluctuations and Horizon Exit

There is now an apparent problem. If inflation makes the universe extraordinarily smooth, why does it not erase the very seeds that galaxies need? The answer is quantum mechanics. The surviving fluctuations are not remnants of an imperfect ironing-out. They are generated anew.

Forget expansion for a moment. Even if a field were almost perfectly homogeneous, quantum mechanics would not let both ϕ and $\dot{\phi}$ be fixed exactly. The ground state is still agitated. This is the same reason an ordinary harmonic oscillator has zero-point energy. In everyday circumstances those fluctuations are hard to see because they are short-wavelength and rapidly oscillating, so observations average over them.

Expansion changes the story. In comoving coordinates the field mode keeps roughly the same coordinate wavelength, while the physical wavelength grows with the scale factor:

$$\lambda_{\text{phys}} = a(t)\,\Delta x. \tag{8.18}$$

During inflation the physical Hubble scale is approximately fixed,

$$d_H \sim H^{-1}, \tag{8.19}$$

so the corresponding comoving horizon shrinks like

$$d_{H,\text{coord}} \sim \frac{1}{aH}. \tag{8.20}$$

The rope analogy makes the freezing intuitive. A wave oscillates because each bit of rope is pulled by neighboring bits. But if the wavelength becomes as large as the horizon, the relevant neighbors are beyond causal contact. The restoring pull disappears. Once a mode is longer than the horizon, it freezes.

This happens again and again. Longer wavelengths freeze first. Shorter wavelengths freeze later. Even a mode that begins with a tiny wavelength is eventually stretched until it crosses the horizon. Thus inflation does not simply erase structure. It keeps producing frozen field variations.

The process also does not stop just because inflation lasts a long time. Old frozen modes are stretched to enormous scales, but new modes are always freezing out. The accumulated pattern becomes statistically steady in the sense relevant for cosmology. That is the origin of the approximately scale-invariant spectrum: the same horizon-crossing process repeats on successive logarithmic scales and produces roughly the same amplitude from one wavelength band to the next.

After the field rolls off the plateau, the horizon grows again. Modes that had been outside the temporary inflationary horizon later come back in. In the usual language they leave the horizon during inflation and re-enter it afterward. By then they are stretched to macroscopic scales, so when they begin to oscillate again they do so as very large structures.

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[a] **Question from the audience.** What does the horizon look like on the spacetime graph?

Response. The physical distance to the horizon stays roughly fixed at c/H during exponential expansion, but on a comoving graph that fixed physical distance looks smaller and smaller because the coordinate scale is being stretched. In comoving terms the horizon shrinks like $1/(aH)$.

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[a] Lecture timestamp: 01:12:02.

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[a] **Question from the audience.** What happens to the amplitude as the fluctuations are stretched, and is there a steady state?

Response. Individual modes dilute as they are stretched, but new modes keep freezing out. The result is a statistical steady pattern, approximately scale invariant, rather than a complete disappearance of fluctuations.

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[a] Lecture timestamp: 01:15:59.

8.6 From Frozen Field Variations to Density Variations

Once the field has acquired small variations from place to place, those variations are converted into density differences when different regions go over the edge at slightly different times. A region a little farther downhill reaches the cliff first. If its vacuum energy is converted into radiation, that radiation dilutes with expansion. A neighboring region still on the plateau remains vacuum dominated and does not dilute. By the time the second region falls, the first one has already redshifted part of its energy away.

That is why slower roll gives larger final density contrasts. If the field moves only very slowly along the plateau, then a small difference in field value corresponds to a larger time lag between one region and the next reaching the cliff. The earlier-falling region has more time to dilute before the later-falling region converts its energy. The flatter the slope, the smaller the terminal velocity, and the larger the final density variation. A higher plateau also tends to give larger fluctuations. The width of the plateau mainly controls how long inflation lasts.

Only a combination of height and slope is tightly constrained by the fluctuation data. If the energy scale is high enough, there should also be a visible gravitational-wave contribution. The numerical bounds are given only as rough ballpark estimates. Roughly speaking, if the energy density on the plateau were much above about 10^{-14} in Planck units, the microwave background would already show too much gravitational-wave structure; if it were much below about 10^{-16}, there would be little realistic chance of seeing such a signal. [4]

The units are explained in the usual natural-units way. The Compton wavelength of a particle of mass m is

$$\lambda_C = \frac{\hbar}{mc}. \tag{8.21}$$

If $c = \hbar = 1$, then mass and inverse length have the same units. Energy density is energy per volume, so it has units of mass to the fourth:

$$[\rho] = (\text{mass})^4. \tag{8.22}$$

Seen at a single point, the accumulation of frozen modes looks like a random walk. Different points in space undergo different random walks, and that is precisely what turns the quantum noise into spatial variation.

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[a] **Question from the audience.** Does ϕ undergo a random walk, and does that make the earlier starting point irrelevant?

Response. At any fixed point the successive frozen modes add with random phase, so ϕ does perform a random walk. That very success is also the difficulty: inflation keeps replacing earlier information with later fluctuations, so the stage before inflation becomes hard to recover.

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[a] Lecture timestamp: 01:33:04.

8.7 What Inflation Hides

The theory is successful in a very specific sense. It correlates fluctuations over a huge range of scales with one underlying mechanism. But the same mechanism washes out what came before. Earlier

[4] Editorial clarification: The speaker explicitly treats these numbers as approximate and says he does not remember the precise values. They are kept here only as order-of-magnitude statements.

irregularities are stretched away, and the quantum fluctuations produced later are the ones that remain relevant. That is why inflation is both attractive and frustrating. It explains the smoothness and the fluctuation spectrum, but it also erases much of the earlier history.

This leads to an uneasiness about the starting point. A very high energy density is not the problem. The problem is that the plateau energy seems neither naturally Planckian nor naturally tiny. It is high in ordinary units, but still small in Planck units. It does not look like an obvious first stopping place for a fundamental theory.

There is also no literal $t = 0$ in this simple picture. Once inflation has lasted too long, almost all information about any earlier stage is gone. If the number of e-foldings were only barely above the minimum needed, perhaps some trace of the previous stage could survive, but in general inflation wipes out the past.

The same caution appears in the treatment of de Sitter space. The pure exponential law is correct for the inflationary branch at late times, but the exact de Sitter solution is better described by

$$a(t) \propto \cosh(Ht). \tag{8.23}$$

That solution contracts to a minimum size and then expands. Whether the contracting branch should be taken literally in cosmology is left open.

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[a] **Question from the audience.** Could there have been no start, with exponential behavior continuing indefinitely into the past?

Response. Not in this simple picture. Exact de Sitter space is not an eternally expanding exponential for all time; it is more like $\cosh(Ht)$, with a contracting branch and an expanding branch. Whether that earlier branch should be taken literally is left as an open question.

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[a]Lecture timestamp: 01:37:31.

8.8 Tuning, Curvature, and Anthropic Bounds

To first approximation the important parameters are the height of the plateau, its slope, and its width $\Delta\phi$. The slope matters because it sets the terminal velocity and therefore the size of the fluctuations. The width matters because it helps determine how long inflation lasts. A common prejudice is that a basic field excursion should not be much larger than the Planck mass, and that the energy density should not exceed order one in Planck units. Under such rough bounds, enough inflation does not fill the whole parameter space. The estimate quoted here is a few-percent tuning, about 2%, to make the slope shallow enough for roughly 60 to 62 e-foldings.

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[a] **Question from the audience.** Can inflation be understood in terms of a parameter space, and how tuned is it?

Response. Yes. To first approximation one thinks in terms of the height, slope, and width of the plateau. The lecture gives a rough estimate that getting enough e-foldings requires tuning at the level of a few percent, about 2%.

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[a]Lecture timestamp: 01:46:02.

That mild tuning is then compared with the much sharper puzzle of the present cosmological constant, of order 10^{-123}. The discussion turns to anthropic questions because both problems can be reframed by asking what is required for structure to form.

A positive cosmological constant acts, in the Newtonian picture, like a universal repulsion proportional to distance. If it were much larger, it would not have to blow galaxies apart today in order to be fatal. It would only have to be large enough to stop the primordial 10^{-5} fluctuations from ever condensing into galaxies. The stated example is that a value roughly two orders of magnitude above the observed one would already do that. This is the point associated with Weinberg's argument: the cosmological constant need not be exactly zero; it need only be small enough to allow structure.

Negative curvature works in the same direction. In the Newtonian language of the earlier lectures, $k < 0$ means recession faster than escape velocity. Too much outward thrust also prevents the small primordial overdensities from ever turning into galaxies. Inflation dilutes that curvature. The quoted lower bound is about 59.5 e-foldings. A value around 62 is then taken as a representative inflationary history safely above the threshold.

This changes the probability question. If one asks for the raw volume of parameter space that gives enough inflation, the answer may be only a few percent. If one conditions on being in a universe where galaxies form and observers exist to ask the question, the conditional probability can become much larger. That is where the discussion closes: not with a final solution, but with the fine-tuning puzzles, the anthropic reinterpretation of them, and the recognition that this is where the great debates remain.